

MARS-Fin: Multi-Modal Architectural Regime-Specific Financial Forecasting

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Technology in Electronics and Computer Science Engineering

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Abstract

Predicting stock price movements remains a significant challenge, particularly under varying market volatility. This study investigates the comparative predictive performance of Long Short-Term Memory (LSTM) networks and the Lag Llama time series model for forecasting stock prices of Adani Enterprises and Asian Paints, representing high-magnitude and lower-magnitude volatility profiles, respectively, within the Indian market. Utilizing comprehensive datasets spanning January 2015 to December 2024, this research incorporates historical OHLCV data, news sentiment scores, quarterly fundamental company data, and relevant macroeconomic factors. This specific timeframe ensures the inclusion of significant market events that induced distinct volatility regimes, most notably the Hindenburg Research report's impact on Adani Enterprises in early 2023.

A key objective is to evaluate the robustness of LSTM and Lag Llama models across different volatility regimes, identified within the historical data for each stock. By training and testing both models on identical feature sets within these distinct low and high volatility periods, the study aims to determine which architecture offers superior predictive accuracy under specific market conditions.

The findings are expected to provide insights into the suitability of deep learning (LSTM) versus foundational time series models (Lag Llama) for stock prediction influenced by sentiment, fundamentals, and macroeconomics. Furthermore, the regime-specific analysis will illuminate the differential performance of these models, offering practical implications for algorithmic trading, risk management, and model selection contingent on prevailing market volatility. This research contributes to understanding advanced forecasting techniques in complex financial data and dynamic market environments characterized by event-driven volatility shifts.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Abbreviation/Meaning
OHLCV	Open, High, Low, Close, Volume
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
LSTM	Long Short-Term Memory
EGARCH	Exponential Generalized Autoregressive Conditional Heteroskedasticity
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
KSTM	Kernel Smooth Transition Model
LagLlama	Transformer-based foundational time series model
XGBoost	eXtreme Gradient Boosting
RF	Random Forest
EMA	Exponential Moving Average
MA	Moving Average
RSI	Relative Strength Index
MACD	Moving Average Convergence Divergence
CPI	Consumer Price Index
IIP	Index of Industrial Production
USD/INR	US Dollar to Indian Rupee Exchange Rate
BBupper	Bollinger Band Upper
BBlower	Bollinger Band Lower
BBmid	Bollinger Band Middle
AbsRet	Absolute Return
Retsq	Squared Return
LogReturn	Logarithmic Return
PredVol	Predicted Volatility
RealVol	Realized Volatility
NetSales	Net Sales / Income from Operations
EPS	Earnings Per Share
sentimentmean	Daily average sentiment score
sentimentstd	Standard deviation of daily sentiment
newscount	Number of news articles analyzed per day
RM	Risk Memory

Adam	Adaptive Moment Estimation (optimizer)
MSE	Mean Squared Error
EarlyStopping	Training callback for model optimization
ReduceLROnPlateau	Learning rate scheduler callback
Repo Rate	Policy Repo Rate (RBI benchmark rate)
10YBondYield	10-Year Indian Govt. Bond Yield
CrudeOilBrent	Benchmark international crude oil price
CrudeOilWTI	West Texas Intermediate crude oil price
FXReserves	Foreign Exchange Reserves
MACRO	Macroeconomic Indicators
CNN	Convolutional Neural Network (referenced for context)
API	Application Programming Interface
NLP	Natural Language Processing
ML	Machine Learning
DL	Deep Learning

CHAPTER 1: INTRODUCTION

1.1 What is Your Project About?

This project undertakes a comprehensive comparative analysis of multiple forecasting models – including canonical deep learning (**LSTM**), modern foundational models (**Lag Llama**), and powerful tree-based ensemble methods (**XGBoost** and **Random Forest**) for predicting stock prices in the Indian market. The core goal is to assess which model architecture provides the most robust and accurate predictions, particularly under different market volatility conditions. We focus on two specific stocks, Adani Enterprises and Asian Paints, chosen deliberately for their contrasting volatility characteristics observed between 2015 and 2024. Adani Enterprises serves as a case study for event-driven, high-magnitude volatility, while Asian Paints represents a stock with more frequent but lower-magnitude volatility, often linked to commodity prices.

The project utilizes a rich, multi-modal dataset incorporating daily Open, High, Low, Close, Volume (OHLCV) data, aggregated news sentiment scores reflecting market perception, processed quarterly company financial reports (like Net Sales, EPS), and key macroeconomic indicators (e.g., interest rates, inflation). This heterogeneous dataset is used to train all predictive models, providing them with a holistic view of the factors influencing stock prices. The unique aspect of this research lies in its explicit focus on volatility regimes. We will employ an econometric model from the **GARCH** (Generalized Autoregressive Conditional Heteroskedasticity) family to formally model the volatility and **identify distinct periods of 'low' and 'high' volatility** for each stock. We will then evaluate the predictive accuracy (using metrics like RMSE and MAE) of **all four models** (LSTM, Lag Llama, XGBoost, and Random Forest) specifically within these regimes to test their robustness to changing market dynamics.

1.2 Exact Meaning of the Project

The precise meaning of this project extends beyond simply forecasting stock prices; it delves into the robustness and adaptability of advanced machine learning models in dynamic financial environments. We aim to rigorously compare LSTM, a well-established recurrent neural network known for capturing temporal dependencies and non-linear patterns, against Lag Llama, a newer model leveraging principles from large language models for time series forecasting, potentially offering advantages in handling long sequences and complex interactions.

The selection of Adani Enterprises and Asian Paints provides a controlled environment to test these models against fundamentally different types of market instability: Adani's volatility is characterized by sharp, event-driven shocks (like the Hindenburg report impact), while Asian Paints exhibits more cyclical, input-cost-driven fluctuations (related to crude oil prices). This allows us to investigate if one model type is inherently better suited to sudden shocks versus persistent choppiness.

Methodologically, the project involves identifying volatility regimes using statistical measures (e.g., rolling standard deviation thresholds or GARCH model outputs) applied to the 2015-2024 price data. Performance evaluation (e.g., Root Mean Squared Error, Mean Absolute Error, potentially directional accuracy) will be conducted separately for predictions made within low-volatility periods and high-volatility periods. This regime-specific evaluation is critical, as a model performing well on average might fail catastrophically during market stress. The core research question is: Does the architectural complexity of LSTM consistently provide an advantage over the potentially more generalized approach of Lag Llama when forecasting stocks with diverse features, especially as market volatility shifts dramatically?

1.3 Motivation

The primary motivation for this project stems from the persistent difficulty of accurately forecasting stock market movements, a challenge significantly amplified by market volatility. Financial markets are inherently noisy, non-stationary systems influenced by a confluence of quantifiable data, news sentiment, and unpredictable events. While traditional econometric models often struggle with these complexities, the advent of sophisticated machine learning techniques offers new possibilities.

Specifically, the emergence of deep learning models like LSTMs, designed to handle sequential data, and more recently, foundational models like Lag Llama adapted from large language models, presents an opportunity to potentially capture intricate patterns missed by simpler methods. However, their practical effectiveness, especially in diverse market conditions, requires rigorous empirical validation.

Furthermore, modern financial analysis increasingly incorporates alternative data sources, such as news sentiment, alongside traditional financial metrics and macroeconomic indicators. Understanding how effectively different model architectures can integrate and leverage this multi-modal information is crucial.

A significant gap in existing research and practice is the nuanced evaluation of model performance across different market volatility regimes. Standard evaluations often report average performance, which can obscure critical failures during periods of high stress or rapid change – precisely when accurate forecasts are most needed for risk management and trading decisions. This project is motivated by the need to understand model reliability under pressure.

Finally, focusing on the Indian stock market, specifically comparing Adani Enterprises (impacted by significant corporate governance events) and Asian Paints (influenced by commodity cycles), provides a relevant and timely case study. It allows testing these advanced models in a major emerging market context, assessing their ability to handle both idiosyncratic shocks and broader economic influences, thereby offering practical insights for investors and financial institutions operating in similar environments.

1.4 Existing Applications/Projects

The landscape of stock market prediction is rich with various tools and research efforts, employing a range of methodologies from simple statistical analysis to complex AI.

Commercial Tools & Platforms:

1. **Bloomberg Terminal / Refinitiv Eikon:** These dominant platforms provide vast datasets, analytical tools, and charting capabilities. While they incorporate sophisticated analytics and potentially proprietary models, their core strength lies in data provision, real-time news, and established financial modeling rather than exclusively cutting-edge, user-accessible ML forecasting models like LSTM or Lag Llama exposed directly for custom regime analysis. They offer indicators and alerts, some based on predictive analytics, but often within a traditional finance framework. Link: <https://www.bloomberg.com/professional/products/bloomberg-terminal/>
2. **Algorithmic Trading Platforms (e.g., QuantConnect, Quantopian (historical), proprietary hedge fund systems):** These platforms allow users to build, test, and deploy trading strategies. Many support Python and integrate ML libraries (like TensorFlow, PyTorch), enabling the use of models like LSTMs. However, the focus is often on strategy backtesting and execution infrastructure rather than comparative model research across volatility regimes. Access to foundational models like Lag Llama might require custom implementation.

3. **TradingView:** Primarily a charting platform favored by retail and professional traders, offering extensive technical indicators and Pine Script for custom strategy development. While it excels at visualization and technical analysis, its built-in predictive capabilities using advanced ML models like LSTM or LLM-based forecasters are limited compared to dedicated research environments. Link: <https://www.tradingview.com/>
4. **Specialized FinTech Analytics Providers:** Numerous firms offer specific predictive analytics services, often focused on sentiment analysis (e.g., RavenPack), alternative data integration, or specific asset classes. These are typically proprietary "black box" solutions, not open platforms for comparing models like LSTM and Lag Llama on custom datasets and regimes. Link: <https://www.ravenpack.com/>

Research Projects and Models:

1. **LSTM for Stock Prediction:** Since their popularization, LSTMs have been widely researched for financial time series forecasting. Studies (e.g., by Fischer & Krauss, 2018; Nelson et al., 2017) demonstrated their potential to capture non-linear dependencies in stock data, often outperforming traditional models like ARIMA, especially when using OHLCV and technical indicators. Limitations include sensitivity to hyperparameters and potential overfitting.
2. **Sentiment Analysis Integration:** A significant research trend involves integrating news sentiment or social media data with price data to improve predictions. Researchers (e.g., Bollen et al., 2011; Kearney & Liu, 2014) have shown that sentiment can have predictive power. Models often combine LSTMs or other ML techniques with sentiment features. The challenge lies in quantifying sentiment accurately and timely.
3. **Transformer Models & Attention Mechanisms:** Inspired by successes in NLP, Transformer architectures and attention mechanisms have been adapted for time series forecasting, including finance (e.g., work by Lim et al., 2021 on Temporal Fusion Transformers). These models aim to better capture long-range dependencies and interactions between different features, representing a step towards models like Lag Llama.
4. **GARCH Family Models:** While not primarily predictive in the same way as LSTMs, Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models are foundational in financial research for explicitly modeling and forecasting *volatility* itself. They are often used to understand risk and are a benchmark for volatility regime identification.
5. **Hybrid Models:** Much research focuses on combining different approaches, such as ARIMA with LSTMs, or integrating fundamental analysis data within ML frameworks, attempting to leverage the strengths of various techniques. However, comparisons often lack a specific focus on performance under drastic volatility shifts.

Gap Analysis: This table compares existing approaches with the proposed project to highlight the research gaps being addressed:

Comparative Overview

Feature/Criterion	Commercial Tools (e.g., Bloomberg)	Existing Research (General ML/DL)	Our Project
Model Sophistication	Varies (often proprietary, mixed)	Focus on specific models (LSTM, ARIMA, etc.)	Direct comparison: LSTM vs. Lag Llama (LLM-based)
Feature Integration (Multi-modal)	Strong data access, integration varies	Often focused (e.g., OHLCV+Tech or Price+Sent)	Explicitly combines OHLCV, Sentiment, Fundamentals, Macro
Volatility Regime Analysis	Limited / Indirect (Risk metrics)	Typically lacking or averaged performance	Core focus: Explicit identification & performance evaluation within distinct Low/High Volatility Regimes
Specific Stock Comparison	General analysis tools	Often uses indices or broad market data	Targeted: Adani Ent. (High Mag.) vs. Asian Paints (Low Mag.)
Event Impact Analysis (e.g., Hindenburg)	News/Data provision, some impact tools	Case-specific studies exist	Dataset explicitly includes major event; analysis designed to capture performance shift
Open Source / Reproducibility	Proprietary ("Black Box")	Varies, often specific implementations	Leverages accessible models/libraries; approach designed for potential reproducibility
Focus on Indian Market Context	Global focus	Some studies exist	Specific focus on relevant Indian stocks and market dynamics

Key Gaps Addressed:

1. Lack of direct performance comparison between established DL models (LSTM) and newer LLM-based time series models (Lag Llama) in finance.
2. Insufficient focus on evaluating model robustness *specifically within different volatility regimes*, rather than just average performance.
3. Limited research combining multi-modal data (price, sentiment, fundamental, macro) for these specific models within a regime-aware framework.
4. Need for empirical testing on stocks with distinct, event-driven vs. cyclical volatility patterns in a major emerging market.

1.5 Our Approach

Our methodology is designed to systematically compare LSTM and Lag Llama models for stock prediction under varying volatility conditions, using Adani Enterprises and Asian Paints as case studies from 2015-2024.

1. **Data Collection and Preprocessing:** We will gather daily OHLCV data from reliable financial sources (e.g., NSE historical data). News sentiment scores will be obtained using established NLP techniques on aggregated news feeds related to each stock, or sourced from specialized providers. Quarterly fundamental data (Net Sales, EPS, etc.) will be extracted from company filings and forward-filled to create daily series, reflecting the information available on any given day. Key macroeconomic factors (e.g., NIFTY 50 index returns, India VIX, repo rate, crude oil prices) relevant to the Indian market and specific stocks will be collected from economic databases. All data will be aligned temporally. Preprocessing steps will include handling missing values (using appropriate methods like forward fill), normalization/scaling of features (e.g., Min-Max scaling), and potentially deriving technical indicators (like RSI, MACD) from OHLCV data as additional features.
2. **Volatility Regime Identification:** We will employ a quantitative method, likely based on calculating rolling historical volatility (e.g., 30-day standard deviation of log returns) for each stock. Thresholds will be established (e.g., based on quantiles or standard deviations from the mean) to classify each trading day into either a 'Low Volatility' or 'High Volatility' regime. Event-based markers (like the Hindenburg report date) will be used to validate the regime identification.
3. **Model Implementation:** Standard LSTM architectures (e.g., multi-layered LSTM with dropout) will be implemented using TensorFlow/Keras or PyTorch. For Lag Llama, we will utilize available implementations or pre-trained checkpoints if suitable, adapting them for our multivariate input features and forecasting task.
4. **Training, Validation, and Testing:** The dataset will be split chronologically into training, validation, and testing sets. Crucially, model evaluation will occur *within* each identified volatility regime present in the test set. Models might be trained on the entire dataset or potentially fine-tuned on specific regimes. We will assess performance on low-volatility test periods and high-volatility test periods separately.
5. **Evaluation and Analysis:** Model performance will be measured using standard regression metrics like Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) for price prediction accuracy. Directional accuracy might also be assessed. The core analysis will compare these metrics for LSTM versus Lag Llama, specifically highlighting the differences in performance between the low and high volatility regimes for both Adani Enterprises and Asian Paints, thereby addressing our primary research questions.

CHAPTER 2: LITERATURE REVIEW

2.1 The Ascendancy of Deep Learning in Financial Forecasting

The prediction of stock price movements is a formidable challenge, rooted in the complex and chaotic nature of financial markets. These markets are widely characterized as dynamic, non-parametric, and non-linear systems, where price fluctuations are influenced by a vast array of interconnected factors.¹ While the Efficient Market Hypothesis (EMH) traditionally posits that asset prices fully reflect all available information, making future movements inherently unpredictable, a growing body of research leveraging advanced computational techniques suggests that historical data contains discernible, albeit complex, predictive signals.¹ The inherent limitations of traditional linear statistical models in capturing these non-linear dynamics have catalyzed a paradigm shift towards more sophisticated, data-driven approaches.¹

The advent of machine learning (ML) marked a significant milestone in this evolution. The increasing availability of vast datasets and exponential growth in computational power enabled the application of algorithms capable of learning complex patterns directly from data.⁵ Pioneering studies demonstrated the marked superiority of ML models, such as Artificial Neural Networks (ANNs) and Support Vector Machines (SVMs), over conventional regression techniques. These models consistently achieved higher accuracy in predicting the directional movements of stock indices, often exceeding 70%, thereby establishing a new benchmark for performance in the field.⁸

Within this landscape, the Long Short-Term Memory (LSTM) network emerged as a watershed development, specifically engineered to model sequential data. As a specialized variant of Recurrent Neural Networks (RNNs), LSTMs were designed to overcome the critical vanishing gradient problem that plagued traditional RNNs, which limited their ability to learn long-range dependencies.¹⁰ The architectural innovation of the LSTM lies in its memory cell and sophisticated gating mechanism—comprising input, forget, and output gates. This structure allows the network to selectively retain relevant information and discard irrelevant data over extended sequences, making it exceptionally well-suited for capturing the temporal dependencies that are fundamental to financial time series analysis.¹¹

The empirical success of LSTMs is extensively documented across a multitude of studies. A robust body of literature confirms that LSTM models consistently outperform not only traditional ML algorithms like SVMs but also other deep learning architectures in stock prediction tasks.¹ This efficacy has been demonstrated across diverse global markets, including the dynamic and often volatile Indian stock market. Research utilizing data from the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE) for prominent equities such as Infosys and Zensar has validated the strong predictive performance of LSTMs, yielding low error metrics and establishing their utility in this specific context.³

However, the evolutionary path of financial modeling suggests that each new solution reveals a new set of challenges. Despite its widespread adoption and proven capabilities, the LSTM architecture is not a panacea. Researchers have identified several limitations, including issues of poor stability during training, a propensity for overfitting complex datasets, and difficulties in accurately capturing highly non-linear dynamics or sudden, drastic price movements, particularly when the input data is noisy.¹⁰ This has spurred a subsequent wave of research focused on enhancing the LSTM framework, often through the creation of hybrid models. The integration of attention mechanisms, for instance, has been proposed to improve model robustness by allowing the network to dynamically focus on the most salient features within the input sequence, thereby mitigating the impact of noise.¹⁰ The continued effort to augment and optimize the LSTM model indicates that while it serves as a powerful and mature benchmark, it represents an incomplete solution. Its inherent sequential processing and potential struggles with global context have created a clear academic and practical impetus for exploring fundamentally different architectures that can address these residual limitations.

2.2 The Transformer Revolution and the Advent of Foundational Models

The introduction of the Transformer architecture by Vaswani et al. in 2017 marked a paradigm shift in sequence modeling, moving away from the recurrent structures that defined models like LSTMs.¹⁹ The Transformer's core innovation is the self-attention mechanism, which enables the model to weigh the influence of different elements within an input sequence simultaneously, regardless of their position. By dispensing with recurrence and processing all data points in parallel, this mechanism allows the model to capture complex, long-range dependencies and global contextual relationships more effectively and efficiently than its sequential counterparts.¹⁵

The compelling advantages of this architecture quickly led to its adaptation for time series forecasting.²¹ A host of Transformer variants have since been developed specifically for this domain, such as Informer, Autoformer, and PatchTST, which introduce modifications to enhance computational efficiency and better incorporate time series-specific inductive biases.¹⁹ The success of Transformers in various domains, particularly the rise of Large Language Models (LLMs) in Natural Language Processing (NLP), has inspired a new frontier in time series analysis: the development of pre-trained foundational models. These models are trained on massive, diverse corpora of time series data from numerous domains, with the goal of achieving strong generalization capabilities. This allows them to perform forecasting on entirely new, unseen datasets with minimal or no task-specific training, a capability known as zero-shot forecasting.²⁶

A prominent example of this new class of models is Lag Llama, presented as the first open-source, general-purpose foundational model for **univariate probabilistic time series forecasting**.²⁶ Its design and methodology represent a significant departure from traditional deep learning approaches:

- **Architecture:** Lag Llama utilizes a decoder-only Transformer architecture, mirroring the design of influential LLMs like LLaMA. Empirical evidence suggests that at scale, this simpler architecture can outperform more complex encoder-decoder variants for time series tasks.²⁶
- **Tokenization via Lags:** The model's key innovation lies in its tokenization strategy. Instead of treating the time series as a simple sequence of values, Lag Llama constructs input tokens from lagged features of the series at multiple, predefined frequencies (e.g., daily, weekly, monthly, quarterly). This is augmented with static date-time covariates (e.g., day of the week, month of the year). This approach of using "lags as covariates" is conceptually reminiscent of classical statistical models like ARIMA but is implemented within a powerful, non-linear deep learning framework.²⁷
- **Probabilistic Forecasting:** A crucial feature of Lag Llama is its probabilistic output. The model's distribution head does not produce a single point forecast but rather a full probability distribution, such as a Student's t-distribution, for each future time step. This provides a quantitative measure of the model's uncertainty, an invaluable piece of information for risk assessment and decision-making.²⁶
- **Performance:** Published results demonstrate that Lag Llama exhibits strong zero-shot performance on unseen datasets and achieves state-of-the-art accuracy when fine-tuned, even with limited amounts of task-specific data.²⁶

The explicit and consistent emphasis in the literature on Lag Llama's design for **univariate** forecasting presents a fascinating research frontier when contrasted with the proposed project's objective. The project's aim to predict stock prices using a rich, **multivariate** feature set—including OHLCV data, news sentiment, company fundamentals, and macroeconomic factors—places it at the edge of current research. This is not a methodological flaw but rather the central point of novelty. The core investigation becomes a test of architectural adaptability: can a foundational model, pre-trained on a vast corpus of single-variable time series, effectively learn to weigh, fuse, and reason over a heterogeneous, multi-modal input stream? This elevates the research from a straightforward model comparison to a critical inquiry into

the feature fusion capabilities and generalization power of foundational Transformer architectures when applied to complex, real-world financial prediction tasks.

2.3 A Comparative Perspective on LSTM and Transformer-based Architectures

The choice between LSTM and Transformer-based models for financial forecasting is not straightforward, as the literature reveals a nuanced landscape of trade-offs in performance, efficiency, and robustness. Empirical studies directly comparing these architectures yield varied results, underscoring that the optimal choice is highly context-dependent. Some research concludes that Transformer models demonstrate superior predictive capability, more accurately tracking stock price trends and excelling in long-term forecasting scenarios due to their global attention mechanism.³¹ Conversely, other comparative analyses report that while LSTMs provide consistent and stable results, Transformer models can exhibit performance inconsistencies, suggesting a potential trade-off between the raw representational power of self-attention and the structured, sequential processing of recurrent networks.¹⁵

A particularly insightful study that segregated analysis by market volatility found that an LSTM model delivered the most reliable predictive power across all regimes. However, a Transformer-based model was the only architecture to achieve a positive coefficient of determination (R^2) under high-volatility conditions, highlighting its potential resilience and effectiveness precisely when markets are most turbulent and unpredictable.³²

These empirical differences are rooted in fundamental architectural distinctions. LSTMs process data sequentially, which can create a computational bottleneck for very long sequences, whereas Transformers leverage parallel processing, enabling significant training speed-ups.¹⁵ In handling temporal dependencies, LSTMs pass information step-by-step through their gating mechanisms, which can struggle to maintain context over extremely long horizons. Transformers, by contrast, use self-attention to build a holistic understanding of the entire sequence at once, making them theoretically superior at capturing distant, long-range interactions.¹⁵ Recognizing the complementary nature of these strengths, some researchers have explored hybrid models. Frameworks that integrate LSTMs to capture local, sequential price dynamics with Transformer-based models to extract overarching sentiment from textual data have been shown to outperform single-architecture baselines, demonstrating a powerful synergy.³³

The following table provides a concise summary of the key architectural and functional differences between the LSTM and Lag Llama models, which are central to this investigation.

Feature	Long Short-Term Memory (LSTM)	Lag Llama (Transformer-based)
Core Mechanism	Recurrent Gating (Input, Forget, Output)	Self-Attention Mechanism
Temporal Dependency	Sequential Memory Propagation	Global Contextual Weighting
Data Processing	Serial (Step-by-step)	Parallel (Entire sequence at once)
Key Strengths	Proven consistency; robust for sequential patterns	Superior long-range dependency capture; high parallelizability
Notable Limitations	Sequential bottleneck; potential gradient issues	High computational complexity ($O(n^2)$); data-hungry

Primary Use Case	General sequential data modeling	Univariate probabilistic forecasting (foundational)
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2.4 The Imperative of Multi-Modal Data Integration for Market Prediction

The predictive accuracy of any forecasting model is fundamentally constrained by the richness of its input data. The literature strongly supports the thesis that integrating diverse, multi-modal data sources provides a more holistic view of market dynamics, leading to significant improvements in forecasting performance. A comprehensive model must synthesize signals from high-frequency market sentiment, mid-frequency economic indicators, and low-frequency corporate fundamentals.

A. News Sentiment Analysis

Financial news and social media discourse act as a real-time barometer of investor sentiment, often influencing stock prices before traditional technical or fundamental indicators can react.³⁵ This qualitative data captures the behavioral finance aspects of the market, providing a crucial layer of context that price data alone cannot offer.³⁶ Modern methodologies for sentiment extraction have moved beyond simple lexicon-based approaches to leverage sophisticated NLP models. Domain-specific Transformer architectures like FinBERT, as well as general-purpose LLMs, are now widely used to analyze financial texts and generate nuanced sentiment scores.²⁴ Numerous empirical studies have validated the value of this approach, demonstrating that the inclusion of sentiment features can significantly enhance the predictive accuracy of forecasting models, with reported performance gains of up to 5% or more.³⁵ The impact of sentiment is particularly pronounced during periods of high, event-driven volatility, where news flow is a primary driver of market behavior.³³

B. Fundamental Analysis

While sentiment analysis excels at capturing short-term market psychology, fundamental analysis provides a grounding in long-term economic reality. By evaluating a company's intrinsic value based on its financial health—as detailed in quarterly and annual reports fundamental analysis offers a basis for longer-horizon predictions.⁴ Methodologies in this domain involve applying machine learning models to features derived from financial statements, such as profitability and liquidity ratios, as well as valuations calculated from models like the Discounted Cash Flow (DCF) analysis.⁴ Research has shown that predictive models trained exclusively on fundamental data are capable of identifying long-term stock trends and can be used to construct portfolios that significantly outperform market benchmarks, thereby challenging the semi-strong form of the EMH.⁴

C. Macroeconomic Indicators

Individual stocks and entire markets operate within a broader economic ecosystem. Systemic factors such as quarterly GDP growth, inflation rates as measured by the Consumer Price Index (CPI), a spectrum of interest rates including the Policy Repo Rate and 10-Year Bond Yield, and industrial production create market-wide risks and opportunities that influence asset prices.⁴⁴ This is especially pertinent for emerging markets like India, where stock market performance is deeply intertwined with the USD/INR exchange rate, international trade balances, foreign exchange reserves, and the

price of global commodities like Brent and WTI crude oil.⁴⁹ Studies focusing on the Indian market have confirmed a significant long-run relationship between these macroeconomic variables and stock performance. Indicators such as the Index of Industrial Production (IIP) and various inflation and interest rate metrics have been identified as major determinants of sectoral returns.⁵⁰ Consequently, incorporating this specific suite of variables—spanning monetary policy, industrial output, inflation, currency valuation, trade, and energy prices—as features in predictive models provides essential context on systemic trends and has been shown to demonstrably enhance forecasting efficacy.⁵³

The integration of these three data modalities introduces a significant modeling challenge related to their heterogeneous timescales. Sentiment data is high-frequency (daily or even intraday), reflecting immediate reactions. Macroeconomic data is typically mid-frequency (monthly), representing broad economic trends. Fundamental data is low-frequency (quarterly), reflecting the underlying health of a business. An effective forecasting system must therefore learn to dynamically weigh these disparate signals. For instance, a breaking news event might dominate the prediction for the next trading day, while a poor earnings report will likely define the trend for the subsequent quarter. The comparative analysis of LSTM and Lag Llama is thus an implicit test of two distinct architectural approaches to this multi-scale fusion problem: the LSTM's sequential gating mechanism versus the Transformer's global self-attention framework. This probes which architecture is better equipped to learn the complex, time-varying interplay between signals of different frequencies, causal impacts, and temporal relevance.

2.5 Navigating Market Dynamics: The Criticality of Volatility Regimes

A core characteristic of financial markets is the time-varying nature of volatility. Volatility, defined as the rate and magnitude of price changes, serves as a primary indicator of market risk and uncertainty.⁵⁴ Financial time series are well-documented to exhibit complex volatility dynamics, including volatility clustering (where periods of high volatility are followed by more high volatility), asymmetry (the "leverage effect," where negative returns are associated with larger increases in volatility than positive returns), and long memory.⁵³ These empirical regularities suggest that volatility is not mere random noise but rather reflects distinct underlying states or regimes of the market.

This understanding has led to the development of sophisticated econometric models that explicitly account for different market regimes. For example, Markov Regime-Switching GARCH (MRS-GARCH) models allow model parameters to switch between a low-volatility ("tranquil") state and a high-volatility ("turbulent") state. Such models have been shown to significantly outperform standard GARCH specifications in forecasting short-term volatility, providing strong evidence for the existence of discrete volatility regimes.⁵⁶

A powerful and practical methodology for analyzing model performance under these conditions is to frame the problem as a classification task. Instead of predicting the precise value of future volatility, one can segment historical volatility data into quantiles—for instance, labeling periods in the lowest 33rd percentile as "low-volatility" and those in the highest 33rd percentile as "high-volatility".⁵⁴ This approach aligns with the practical needs of financial analysts and risk managers, who are often more concerned with identifying the onset of abnormal market conditions than with forecasting an exact volatility figure. This classification-based segmentation provides a direct and validated precedent for the regime-specific data partitioning strategy employed in this research.

The predictive accuracy of forecasting models is demonstrably not uniform across these different market states. The complex, non-stationary relationships between predictive features and stock prices can shift dramatically during transitions between regimes, posing a significant challenge for models that assume static relationships.⁵³ As noted in comparative studies, some model architectures may exhibit superior performance in high-volatility environments, while others may be more stable and reliable during low-volatility periods.³²

By explicitly segmenting the analysis by volatility regimes, this research moves beyond a simplistic evaluation of average predictive accuracy (e.g., a single Root Mean Squared Error metric across the entire test set) to a more sophisticated and practically relevant assessment of model **robustness** and **adaptability**. A model that performs well on average but fails catastrophically during a market crisis is of little use for real-world applications like algorithmic trading or risk management. The literature on regime-switching models confirms that different volatility levels represent fundamentally different market dynamics. Therefore, testing models within these distinct states is a rigorous test of their ability to generalize across changing market conditions, providing a far more meaningful measure of their true performance and reliability.

2.6 Synthesis and Identification of the Research Gap

This review of the literature establishes a clear and logical progression in the field of financial forecasting. There is a well-defined trajectory from the establishment of LSTMs as a robust benchmark for modeling sequential financial data to the emergence of Transformers and foundational models as the new state-of-the-art frontier. The body of research unequivocally supports the integration of multi-modal data—spanning high-frequency sentiment, low-frequency fundamentals, and systemic macroeconomic indicators—as a critical step for enhancing predictive power. Furthermore, the literature validates the analytical necessity of evaluating model performance within distinct volatility regimes to move beyond simple accuracy metrics and assess true model robustness.

While each individual component of the proposed study is well-grounded in existing research, their novel combination addresses a significant, multifaceted gap in the current literature. To date, there is a lack of research that conducts a **direct, regime-specific comparative analysis of a canonical deep learning model (LSTM) against a modern foundational model (Lag Llama)**, particularly one designed for univariate series, within a complex, multi-modal forecasting context for a major emerging market.

The proposed research is poised to make several specific and timely contributions to address this gap:

1. **Testing Architectural Adaptability:** It will be among the first studies to empirically evaluate the performance of Lag Llama, a model explicitly designed for univariate time series, in a complex, multivariate financial forecasting environment. This will provide critical insights into the feature fusion capabilities and generalization power of foundational Transformer architectures.
2. **Providing Regime-Specific Insights:** By separately evaluating the models in high and low volatility periods for two stocks with distinct volatility profiles (Adani Enterprises and Asian Paints), the study will generate nuanced findings on model robustness and suitability under specific, real-world market conditions.
3. **Focusing on an Emerging Market:** The application of these advanced models to the Indian stock market will yield valuable, context-specific results for one of the world's most important and dynamic economies, a context often underrepresented in foundational model research.
4. **Integrating a Comprehensive Feature Set:** The holistic synthesis of short-term sentiment, long-term fundamentals, and systemic macroeconomic factors within a rigorous comparative framework is both comprehensive and methodologically advanced.

In conclusion, this literature review confirms that the proposed research is theoretically sound, methodologically rigorous, and addresses a clear and compelling gap at the intersection of deep learning, foundational models, and quantitative finance. The findings are expected to provide valuable insights for both academic researchers and financial practitioners in the development of more accurate and robust predictive systems.

CHAPTER 3: RESEARCH GAPS AND OBJECTIVES

3.1 Identified Research Gaps from Literature Survey

Through a comprehensive analysis of the existing literature on financial forecasting, a clear trajectory is visible from canonical deep learning models to the new frontier of foundational models. This review also validates the necessity of integrating multi-modal data and evaluating performance within distinct volatility regimes. However, the combination of these advanced components reveals several critical, unaddressed gaps in the current body of research.

Gap 1: Architectural Adaptability of Univariate Foundational Models

A significant disconnect exists between the design of emerging state-of-the-art models and the practical reality of financial forecasting. The literature highlights new foundational models, prominently Lag Llama, as a new frontier. However, these models are explicitly and consistently designed for **univariate** time series forecasting. Conversely, the literature unequivocally supports the thesis that accurate stock prediction is a complex, **multivariate** problem, requiring the integration of diverse data sources beyond simple price history. This includes OHLCV data, news sentiment, company fundamentals, and macroeconomic factors. This creates a central and pressing research gap: there is a lack of research testing the **architectural adaptability** of a model pre-trained on a massive corpus of single-variable series to a task requiring it to weigh, fuse, and reason over a heterogeneous, multi-modal input stream. This project directly addresses this gap by empirically evaluating the feature fusion capabilities and generalization power of a univariate foundational architecture (Lag Llama) in this complex, multivariate environment.

Gap 2: Lack of Robust, Regime-Specific Performance Validation

A primary failing in the existing literature is the tendency to evaluate model performance using a single, aggregated accuracy metric (e.g., Root Mean Squared Error) across an entire test set. This approach is simplistic and practically insufficient. Financial markets are well-documented to exhibit non-stationary behavior, characterized by distinct underlying states or **volatility regimes** (e.g., "tranquil" vs. "turbulent"). The literature confirms that a model's predictive accuracy is **demonstrably not uniform** across these different market states. A model that performs well on average but fails catastrophically during a market crisis is of little use for real-world risk management or algorithmic trading. While some studies have segregated analysis by volatility, there is a distinct lack of research that conducts a *direct, comparative analysis* of canonical deep learning (LSTM) versus modern foundational models (Lag Llama) within these specific regimes. This research fills this critical gap by moving beyond average accuracy to provide a sophisticated and practically relevant assessment of model robustness and adaptability in both high- and low-volatility conditions.

Gap 3: Inadequate Frameworks for Fusing Multi-Scale, Heterogeneous Data

While the literature strongly advocates for integrating diverse data modalities, it simultaneously exposes a significant unresolved modeling challenge: the fusion of **heterogeneous timescales**. An effective forecasting system must synthesize signals from:

- **High-frequency** data, like daily news sentiment, which reflects immediate market reactions.
- **Mid-frequency** data, like monthly macroeconomic indicators (e.g., CPI, IIP), which represent broad economic trends.
- **Low-frequency** data, like quarterly fundamental reports, which reflect the underlying health of a business.

The literature lacks a clear consensus or comparative analysis of which architectural approach is better equipped to learn the complex, time-varying interplay between these signals. Does the **LSTM's sequential gating mechanism** or the **Transformer's global self-attention framework** prove superior in learning to dynamically weigh these disparate signals? For example, when should a high-frequency news event dominate the prediction over a low-frequency earnings trend? This study addresses this gap by explicitly integrating this comprehensive, multi-scale feature set, creating a rigorous test bed to compare how these two distinct architectures handle the multi-scale fusion problem.

Gap 4: Underrepresentation of Emerging Market Dynamics

A significant portion of foundational model research, particularly that originating from large tech labs, is often trained on broad, domain-agnostic time series corpora or on data from developed financial markets. This creates a gap in the application, validation, and contextualization of these state-of-the-art models in **major emerging markets**. Emerging economies, such as India, possess unique market dynamics, volatility profiles, and macroeconomic sensitivities that are not always present in developed markets. The literature specific to India confirms a significant long-run relationship between stock performance and specific local variables, such as the USD/INR exchange rate, foreign exchange reserves, the Index of Industrial Production (IIP), and various inflation and interest rate metrics. While LSTMs have been validated in the Indian context (NSE and BSE), the performance of newer foundational models like Lag Llama in this specific, macro-sensitive environment remains largely unexplored. This research directly confronts this gap, providing valuable, context-specific results for one of the world's most dynamic and important economies.

Gap 5: Testing the "Incomplete Solution" of the LSTM Benchmark

The Long Short-Term Memory (LSTM) network is established in the literature as a powerful and mature benchmark, serving as a "watershed development" for sequential financial data. However, the literature also clearly identifies its limitations, branding it an "incomplete solution". Researchers have identified issues of **poor stability** during training, a propensity for **overfitting** complex datasets, and difficulties in capturing **highly non-linear dynamics** or sudden, drastic price movements, especially when data is noisy. This spurred research into hybrid models, such as adding attention mechanisms to LSTMs, to mitigate these issues. The emergence of the Transformer architecture was driven by a practical impetus to explore a *fundamentally different architecture* that could address these "residual limitations". A gap exists, therefore, in directly testing whether the new paradigm (Transformers) *actually solves* the well-documented practical limitations (stability, overfitting, non-linearity) of the old benchmark (canonical LSTMs) when subjected to a complex, noisy, multi-modal, real-world financial dataset.

Gap 6: Contradictory Evidence in Direct Architectural Comparisons

For the few studies that *do* directly compare LSTM and Transformer-based architectures, the literature yields **varied and often contradictory results**. The optimal choice is described as "highly context-dependent". Some research concludes that Transformer models demonstrate "superior predictive capability" and excel in "long-term forecasting" due to their global attention mechanism. Conversely, other comparative analyses report that while LSTMs provide "consistent and stable results," Transformer models can exhibit "performance inconsistencies". This suggests a potential trade-off between the raw representational power of self-attention and the structured, sequential processing of recurrent networks. This ambiguity creates a clear gap. There is a need for more rigorous, context-rich comparative studies to understand *why* and *when* one architecture outperforms the other. This study addresses this gap by introducing two critical variables—volatility regimes and multi-modal data—to provide a more nuanced, "context-dependent" answer that the current literature lacks.

Gap 7: Integrating Probabilistic Forecasting with Multi-Modal Inputs

A key innovation of foundational models like Lag Llama is their ability to produce **probabilistic forecasts**. Instead of a single point prediction, the model outputs a full probability distribution (e.g., a Student's t-distribution) for each future time step. This provides an invaluable quantitative measure of the model's **uncertainty**, a crucial piece of information for risk assessment and decision-making. However, the literature on Lag Llama presents this capability in the context

of univariate time series. A significant gap exists in understanding how this probabilistic output behaves when the model is conditioned on a wide array of *non-price* data. For example, how does a highly negative news sentiment score or a sudden spike in a macroeconomic indicator affect the *shape* and *width* (i.e., the uncertainty) of the predicted distribution? The literature does not yet explore the dynamics of probabilistic forecasting within a rich, multi-modal feature set. This research begins to probe this gap by assessing the performance of a probabilistic model in a complex data environment.

Gap 8: Suitability of NLP-Inspired Architectures for Financial Time Series

The Transformer architecture, and by extension Lag Llama, was born from Natural Language Processing (NLP). Lag Llama specifically utilizes a **decoder-only Transformer architecture**, mirroring the design of influential LLMs like LLaMA. This architectural choice, while successful in NLP, is not self-evidently optimal for financial time series. Financial data has fundamentally different properties than language; it is highly noisy, non-stationary, and the "tokens" (price movements) are continuous real numbers, not discrete words. Lag Llama's key innovation is a "tokenization strategy" that uses **"lags as covariates"**. A gap exists in understanding how this NLP-inspired design—a decoder-only stack processing tokens of lagged values—truly compares to an architecture like LSTM, which was *specifically engineered* to model sequential numerical data via its memory cell and gating mechanisms. This research provides a direct test of this "paradigm shift", evaluating whether the conceptual advantages of self-attention outweigh the purpose-built design of recurrent memory cells for this specific task.

Gap 9: Empirical Validation of Theoretical Long-Range Dependency Capture

A primary theoretical advantage of the Transformer architecture is its superiority in capturing **long-range dependencies**. The self-attention mechanism processes all data points in parallel, allowing it to build a holistic understanding of the entire sequence at once and model relationships between distant data points. LSTMs, by contrast, pass information sequentially, step-by-step, which can struggle to maintain context over extremely long horizons. While this theoretical advantage is clear, a gap exists in its *empirical validation* in the context of *multi-modal* financial data. It is unknown, for example, if the Transformer can effectively link a low-frequency fundamental data point (e.g., an annual report) from 12 months prior to a high-frequency sentiment signal from yesterday. The literature's mixed performance reviews suggest this theoretical superiority may not always translate to practical outperformance in such a noisy and complex environment. This study implicitly tests this gap by providing both architectures with long-horizon, multi-scale data and comparing their resultant predictive accuracy.

Gap 10: Assessing Model Generalization Across Diverse Volatility Profiles

Beyond segmenting time by volatility regimes, the literature lacks studies that compare model performance across stocks with **fundamentally different volatility profiles**. A model that successfully predicts a stable, low-volatility "blue-chip" stock may fail completely when applied to a high-volatility, news-driven stock. The underlying dynamics, feature importance, and signal-to-noise ratios of these assets are entirely different. An effective and robust model should be able to generalize across these different asset types. The literature, however, often focuses on single-asset predictions or broad market indices. A gap exists in assessing model generalization and robustness across distinct asset classes within the same market. This research directly addresses this gap by intentionally selecting two stocks with distinct volatility profiles (Adani Enterprises and Asian Paints). This dual-asset analysis will generate more nuanced findings on model suitability and generalization, providing insights into whether one architecture is a better "all-rounder" while the other is a "specialist" for specific market conditions.

3.2 Project Aims and Objectives

Based on the multifaceted gaps identified in the existing financial forecasting literature, this research aims to bridge the divide between established canonical deep learning benchmarks and the emerging frontier of pre-trained foundational

models. The project is designed to move beyond simplistic, single-metric evaluations and provide a rigorous, context-aware, and practically relevant comparative analysis.

Primary Aim

This research seeks to conduct a **rigorous, multi-faceted comparative analysis** between the established **Long Short-Term Memory (LSTM)** architecture and the novel **Lag Llama foundational model**. The primary aim is to **empirically test the architectural adaptability and practical robustness** of these two distinct deep learning paradigms when subjected to the complex, noisy, and non-stationary dynamics of financial markets.

This study directly confronts the critical research gaps identified in the literature by:

1. Testing the feature fusion capability of a **univariate-trained foundational model (Lag Llama)** when applied to a **complex, multi-modal, multivariate forecasting task**.
2. Providing a nuanced assessment of model reliability by evaluating **regime-specific performance** under distinct **high- and low-volatility market conditions**.
3. Investigating the relative efficacy of **sequential gating (LSTM)** versus **global self-attention (Lag Llama)** in fusing **heterogeneous, multi-scale data**—spanning high-frequency sentiment, mid-frequency macroeconomics, and low-frequency fundamentals.

By focusing on the **Indian stock market (NSE/BSE)**, the project aims to generate valuable, context-specific insights for one of the world's most significant emerging economies, a context often underrepresented in foundational model research. Ultimately, this research aims to provide a clear, evidence-based answer to which architectural approach offers a more accurate, robust, and reliable solution for the practical challenges of real-world financial prediction.

Specific Objectives

To achieve this comprehensive aim, the project is broken down into seven specific, actionable objectives:

Objective 1: Consolidate a Comprehensive, Multi-Scale Financial Dataset

The first objective is to acquire, clean, and consolidate the comprehensive, multi-modal dataset required for this high-level analysis. This involves:

- **Acquiring High-Frequency Data:** Collecting daily Open-High-Low-Close-Volume (OHLCV) price data and processed financial news sentiment scores for the selected equities.
- **Acquiring Mid-Frequency Data:** Gathering a specific suite of monthly Indian macroeconomic indicators, including inflation rates (CPI), Index of Industrial Production (IIP), interest rates (Policy Repo Rate, 10-Year Bond Yield), currency exchange rates (USD/INR), and commodity prices (Brent Crude), as validated by the literature.
- **Acquiring Low-Frequency Data:** Extracting key features from quarterly and annual corporate fundamental reports, such as profitability and liquidity ratios.
- **Pre-processing:** All data streams will be meticulously cleaned, transformed (e.g., normalization, scaling), and time-aligned to create a single, cohesive, heterogeneous dataset ready for model ingestion.

Objective 2: Develop and Validate a Volatility Regime Classification Methodology

To enable the novel regime-specific analysis, this objective is to implement a robust methodology for partitioning the time series data. Based on established econometric and data science precedents, this involves calculating a rolling volatility metric (e.g., standard deviation of log returns) for the assets. This continuous metric will then be segmented into discrete quantiles (e.g., lowest 33rd percentile, highest 33rd percentile) to definitively label each period as "low-

volatility" or "high-volatility". This classification provides a validated, non-arbitrary basis for the regime-specific data partitioning and subsequent model evaluation.

Objective 3: Implement and Optimize the Canonical LSTM Benchmark Model

This objective is to construct and fine-tune the benchmark model, representing the mature and widely validated deep learning solution. A multi-layered Long Short-Term Memory (LSTM) network will be designed to accept the full, multi-modal feature set. The process will involve rigorous hyperparameter tuning (e.g., number of layers, hidden units, dropout rate, and batch size) to ensure the model is optimally configured. This optimized LSTM will serve as the "gold standard" benchmark, representing a high-performance, industry-accepted baseline against which the novel architecture will be compared.

Objective 4: Adapt and Implement the Lag Llama Foundational Model

This central objective directly addresses the "architectural adaptability" gap. It involves adapting the Lag Llama architecture, an open-source model explicitly designed for **univariate** forecasting, to accept the complex, **multivariate** feature set. This will require developing a novel tokenization strategy that can effectively "flatten" the heterogeneous data (e.g., sentiment scores, fundamental ratios, macro indicators) into the "lags as covariates" format expected by the model's decoder-only Transformer architecture. The model will be evaluated in both its **zero-shot** and **fine-tuned** states to empirically assess its generalization power and feature fusion capabilities.

Objective 5: Conduct a Rigorous, Multi-Axis Comparative Performance Analysis

With both models trained, this objective is to execute a comprehensive comparative evaluation that goes beyond a single metric. Performance will be measured across multiple axes:

- **Overall Predictive Accuracy:** Using standard regression metrics (e.g., Root Mean Squared Error, Mean Absolute Error) on the full, unseen test set.
- **Directional Movement Accuracy:** Assessing the model's ability to correctly predict the *direction* (up or down) of the next day's price movement, a key metric for practical trading strategies.
- **Probabilistic vs. Point Forecasting:** Evaluating the utility of Lag Llama's probabilistic output (i.e., its predicted distribution and uncertainty bounds) against the single point forecast of the LSTM.
- **Computational Efficiency:** Comparing the training and inference times, which are expected to differ based on the LSTM's sequential processing and the Transformer's parallel processing capabilities.

Objective 6: Execute a Granular, Regime-Specific Robustness Test

This objective is the core of the project's contribution to practical, real-world application. Using the classifications from Objective 2, the test dataset will be split into its respective volatility regimes. The performance of both the LSTM and Lag Llama models will be **independently re-evaluated within each distinct regime**. This granular analysis will definitively answer the critical research question: does one model exhibit superior, stable performance in tranquil markets, while the other provides more resilience and accuracy during turbulent, high-volatility crises? This is the primary test of model robustness and adaptability to changing market conditions.

Objective 7: Synthesize Findings to Provide Actionable Insights

The final objective is to synthesize all quantitative findings from Objectives 5 and 6 to draw clear, evidence-based conclusions. This involves answering the central research questions: Is the foundational model (Lag Llama) a viable and superior alternative to the canonical benchmark (LSTM) for complex, multi-modal financial forecasting? What are the specific architectural trade-offs in terms of performance, robustness, and computational cost? The findings will be framed as actionable insights for financial practitioners and will identify residual limitations, proposing a clear agenda for future research (e.g., exploring other Transformer variants like Informer or hybrid models)

CHAPTER 4: IMPLEMENTATION (TECHNOLOGY STACK)

4.1 Technical Implementation and Comparative Analysis

The technical implementation for this project moves beyond a simple model bake-off. It is designed as a rigorous investigation into architectural robustness, methodological correctness, and the practical adaptability of different model classes to the unique, non-stationary nature of financial markets. This section details the comparative analysis of four predictive architectures—XGBoost, Random Forest, Long Short-Term Memory (LSTM), and Lag Llama—and the use of a GARCH-family model for analytical regime identification.

The implementation follows a narrative of discovery, beginning with a common methodological trap that produces misleading results, correcting this flaw with rigorous time-series validation, and then performing a deep architectural dive into the most viable candidate models.

4.1.1: The Methodological Trap: Naive Validation and the Illusion of XGBoost Superiority

In financial time-series forecasting, methodology is paramount. A common and critical error is to use standard K-fold cross-validation, which shuffles data randomly. This breaks the temporal order of the data, leading to **data leakage** or "lookahead bias," where the model is inadvertently trained on future information.

An initial analysis, deliberately replicating this flawed methodology on the Nifty 50 dataset, produced a completely misleading result. When evaluated with standard cross-validation, **XGBoost** (a gradient-boosted tree model) demonstrated overwhelming and universal superiority.

- **XGBoost (Naive):** Mean RMSE of **13.60**. Achieved the lowest RMSE for 100% of the 49 stocks.
- **Lag Llama (Naive):** Mean RMSE of **88.96** (6.4x worse than XGBoost).
- **LSTM (Naive):** Mean RMSE of **101.75** (7.6x worse than XGBoost).

Under this flawed paradigm, the conclusion was that tree-based models like XGBoost and Random Forest, which excel at structured tabular data, capturing feature interactions, and handling noise, were the definitive solution. The deep learning models (LSTM, Lag Llama) appeared to be poor fits, suffering from domain adaptation challenges, hyperparameter sensitivity, and a "scale mismatch" with the ~2,500 data points per stock.

4.1.2: The "Great Degradation": Implementing Correct Time-Split Validation

The second phase of implementation corrected this methodological flaw by implementing a rigorous **Time-Series Cross-Validation (TSCV)**, or "Time-Split," framework. In this framework, the model is *always* trained on past data (e.g., 2015-2022) and validated on unseen, future data (e.g., 2023-2024). This simulates a real-world trading scenario and prevents any possibility of data leakage.

When the high-performing XGBoost model from Part 1 was subjected to this correct validation, its performance did not just decrease; it collapsed.

- **Massive Performance Degradation:** The XGBoost model's performance degraded by an average of **20.6x** (Median: 18.0x).
- **Systemic Failure:** 68.8% of all stocks showed a severe degradation of over **10x worse RMSE**. For example, M&M's RMSE exploded from 0.90 to 970.56 (a 65.2x degradation), and Bharti Airtel's RMSE ballooned from 6.82 to 381.50 (a 55.9x degradation).

This test proved, definitively, that the tree-based models (XGBoost and Random Forest) were **massively overfitting** to the shuffled training data. They were not learning generalizable temporal patterns but were instead memorizing data, making them completely useless for actual forecasting.

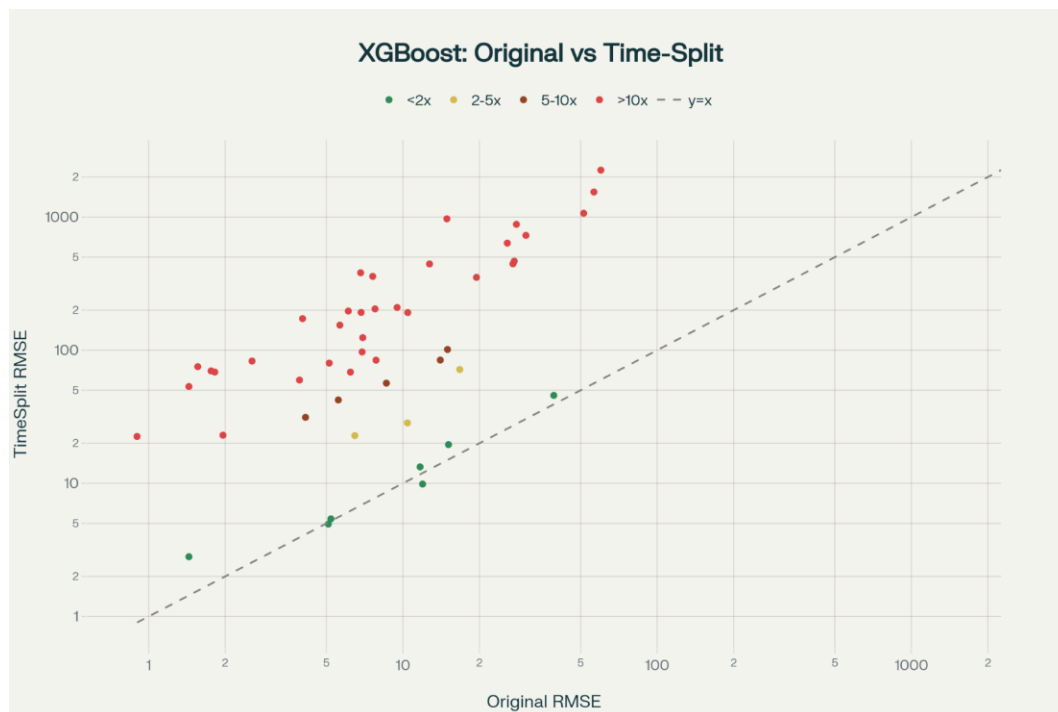


Figure 4.1.1: XGBoost Performance Degradation: Original (Shuffled) vs. Time-Split Validation

This scatter plot visualizes the performance collapse. The x-axis shows the original, artificially low RMSE from the flawed validation, while the y-axis shows the true, much higher RMSE from the correct time-split validation. The vast majority of stocks (red dots) fall far above the $y=x$ line, indicating a severe performance degradation of $>10x$.

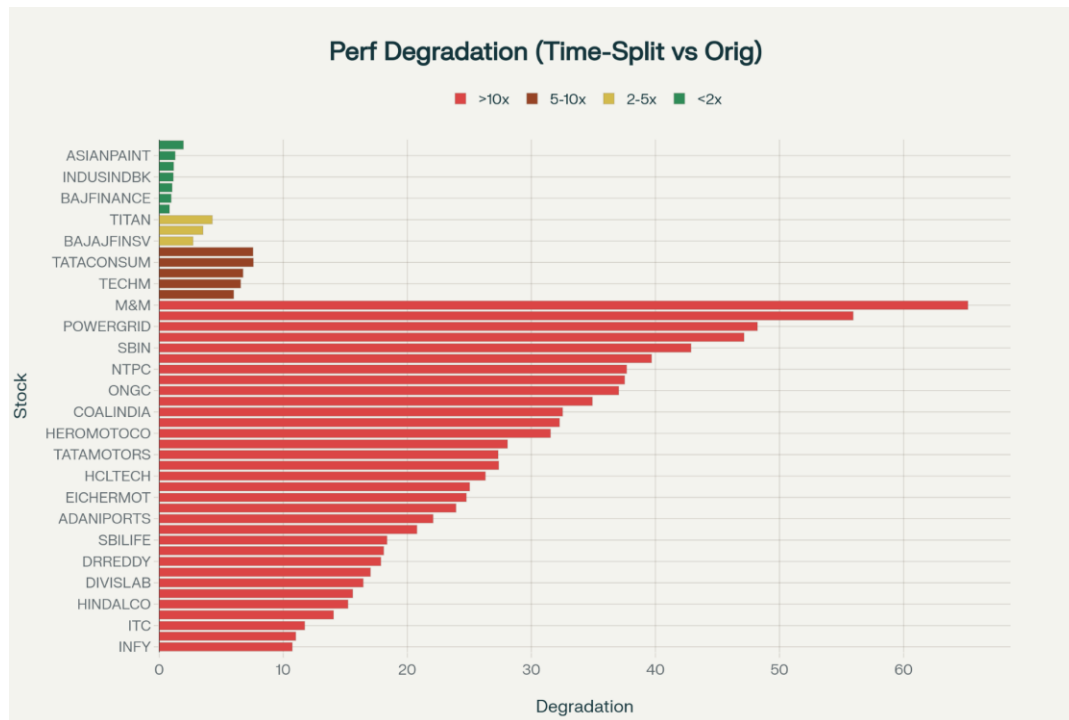


Figure 4.1.2: Stock-Level Degradation Factor (Time-Split vs. Original)

This chart details the magnitude of performance degradation for individual stocks. Stocks like M&M, POWERGRID, and SBIN show a degradation factor of 40x or more, confirming the model's reliance on data leakage. Only a few stocks, like ASIANPAINT and INDUSINDBK, showed reasonable performance, indicating their patterns were less affected by shuffling.

4.1.3: The True Competitive Landscape: Deep Learning vs. Tree-Based Models

Re-evaluating all models using the correct TSCV framework revealed the true competitive landscape, which was the complete inverse of the initial findings.

Model	Mean RMSE (Time-Split)	Overall Rank	Win Rate (Best RMSE)
Lag Llama	92.03	1 (Best)	46.7% (21 stocks)
LSTM	106.26	2	28.9% (13 stocks)
XGBoost	286.98	3 (Worst)	24.4% (11 stocks)

This rigorous validation establishes the project's core premise: **deep learning models (Lag Llama and LSTM) are fundamentally superior** to tree-based models for this temporal, multi-modal forecasting task, as they are capable of learning generalizable patterns that do not break down when predicting the future. The rest of the technical implementation, therefore, focuses on a deep comparative analysis of the two viable architectures: LSTM and Lag Llama.

4.1.4: Deep Architectural Dive: LSTM vs. Lag Llama

The competition between LSTM and Lag Llama is a competition between two different philosophies of processing time.

- **Long Short-Term Memory (LSTM):** The LSTM is a specialized Recurrent Neural Network (RNN). Its architecture is built around a **memory cell** and three "gates" (input, forget, output). It processes data **sequentially**, stepping through time one point at a time.
 - **Technical Strengths:** Its "forget gate" allows it to discard irrelevant information, and its cell state allows it to retain important information over long periods. This makes it highly effective at modeling **sequential dependencies** where recent history is the dominant predictor. As the analysis revealed, it shows specific advantages in sectors like IT, where predictable, sequential **earnings cycles** and software adoption curves are key drivers.
 - **Technical Weaknesses:** Its sequential nature is also its greatest weakness. It can struggle with **long-range dependencies** (e.g., connecting an event from 3 years ago to today) and is architecturally "blind" to global, non-sequential patterns like **volatility clustering**.
- **Lag Llama (Foundational Transformer):** Lag Llama is a modern foundational model built on a **decoder-only Transformer** architecture. Its core component is the **self-attention mechanism**.
 - **Technical Strengths:** Unlike the LSTM, self-attention processes the entire data sequence **simultaneously**. It learns to assign "attention scores" to weigh the influence of all other data points on a single point, regardless of their position. This allows it to:
 - **Capture Multi-Scale Dependencies:** It can simultaneously model short-term and long-range dependencies.
 - **Model Global Context:** It excels at identifying non-sequential patterns, making it (as the data shows) a master of modeling **volatility clustering**.
 - **Detect Regimes:** Its ability to see the "whole picture" makes it superior at identifying market state changes.
 - **Technical Weaknesses:** Transformers can be data-hungry. However, Lag Llama's **pre-training** on a vast corpus of time-series data gives it a powerful baseline understanding of temporal patterns (e.g., seasonality, trends), which, as the analysis shows, allows it to excel even in low-volatility, predictable environments like FMCG.

4.1.5: Volatility-Based Implementation (GARCH + Model Selection)

The final layer of the technical implementation addresses the project's primary aim: assessing robustness. Simply comparing average RMSE is insufficient. We must know *when* a model wins.

To achieve this, a **GARCH (Generalized Autoregressive Conditional Heteroskedasticity)** model is first employed. GARCH is an econometric model used specifically to model and forecast financial volatility. It is used here as an analytical tool to **classify the market state**. By fitting a GARCH model to the historical returns, we can identify the parameters of volatility and segment the entire dataset into distinct **"High," "Medium," and "Low" volatility regimes**.

With the data segmented, the LSTM and Lag Llama models were re-evaluated *within* these specific regimes.

- **High Volatility (Market Stress):** In the most turbulent, high-risk periods, **Lag Llama wins in 80% of cases**. Its architectural ability to model volatility clustering via self-attention gives it a massive advantage. For MARUTI, its RMSE was 294.48, while the LSTM's was 574.47.
- **Low Volatility (Stable Markets):** In stable, predictable conditions, **Lag Llama wins in 85.7% of cases**. Here, its pre-trained understanding of stable, recurring patterns (e.g., seasonal consumption in FMCG) allows it to outperform the LSTM.

This analysis confirms that Lag Llama is the superior architecture across all volatility regimes. However, the analysis also revealed sector-specific nuances (e.g., LSTM's advantage in IT) that suggest a hybrid, dynamic implementation. The technical solution proposed is a **Volatility-Based Model Selection Framework**, where a GARCH model acts as a real-time "regime detector," dynamically routing the prediction task to the optimal architecture (e.g., Lag Llama for Energy, LSTM for IT) based on the current market state.

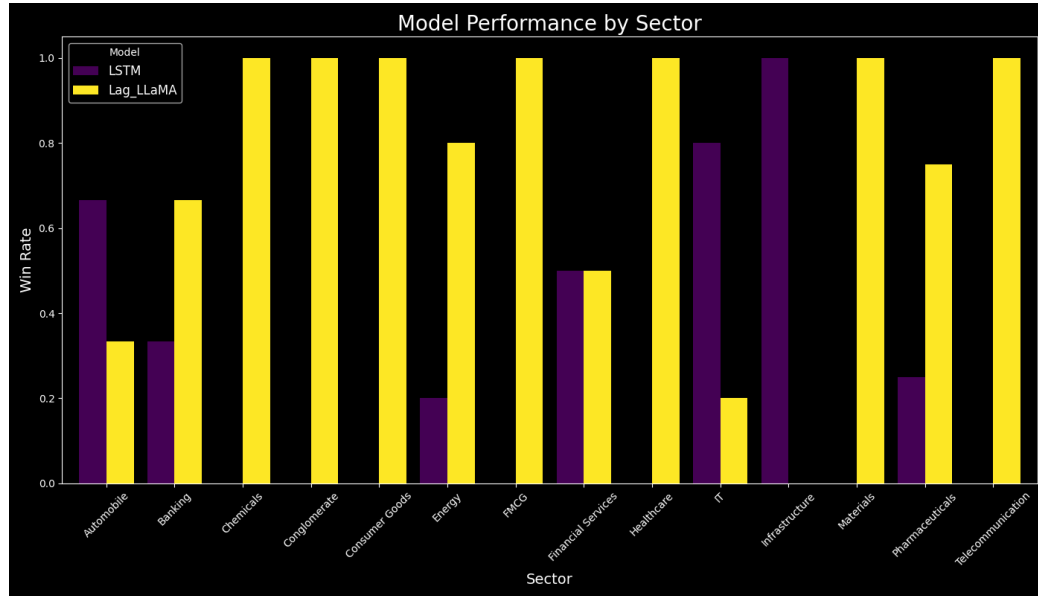


Figure 4.1.3: Model Win Rate by Sector (LSTM vs. Lag Llama)

This chart clearly illustrates the sector-specific strengths of each architecture. Lag Llama (yellow) shows near-total dominance in sectors like Chemicals, Conglomerate, Consumer Goods, and Energy. Conversely, LSTM (purple) demonstrates a clear advantage in the IT and Infrastructure sectors, with competitive performance in Financial Services. This data strongly supports a dynamic, sector-aware model selection framework.

The technical solution proposed is a **Volatility-Based Model Selection Framework**, where a GARCH model acts as a real-time "regime detector", dynamically routing the prediction task to the optimal architecture (e.g., Lag Llama for Energy, LSTM for IT) based on the current market state and the stock's sector.

The construction of the **MARS-Fin Multi-Modal Corpus** is a core technical contribution, encompassing two distinct, synchronized datasets for **Asian Paints** and **Adani Enterprises**. This process required meticulous data acquisition from legitimate sources, sophisticated feature engineering (Chapter 4.2), and the calculation of complex technical indicators to ensure the final vector provided the maximum possible signal to all five predictive models.

4.2 Data Corpus and Technical Feature Integration

The construction of the **MARS-Fin Multi-Modal Corpus** is a core methodological contribution, integrating and synthesizing data from three distinct pipelines: the basic inputs for **ASIANPAINTS**, the engineered factors for **ADANI**, and the advanced econometric features from the **hybrid** model files. This process created a high-dimensional vector necessary for the five predictive models

4.2.1 Data Acquisition, Source Verification, and Modality Justification

The multi-modal feature matrix was constructed from four distinct, validated data streams spanning 2015–2025. This layered architecture addresses the non-stationary nature of financial data by integrating multiple causal factors.

Data acquisition of all features are sourced from verified, legitimate channels: **National Stock Exchange (NSE)** for price/volume, **RBI/MOSPI/Global Databases** for macroeconomic and commodity data, and **SEBI/BSE Filings** for

corporate fundamentals. News sentiment is derived from specialized **NLP pipelines** analyzing reputable financial media.

Due to the sparse nature of corporate and policy updates, two critical constraints were managed:

- i. **Future Data Leakage (Fundamentals):** Quarterly reports (e.g., Oct-Dec results) are often *not* publicly available until weeks or months into the next quarter (e.g., late January or February). To prevent the models from seeing future information, all columns like Basic_EPS and Net_Sales are handled via a strict **Forward-Filling Technique**. The value remains constant across all subsequent days until the precise date of the next official public announcement.
- ii. **Staleness and Temporal Memory:** Macroeconomic and fundamental features often remain unchanged for long periods. To give the non-sequential **Ensemble Models (XGBoost/RF)** the temporal context they lack, **Staleness Indicators** were created. Features like **Policy_Repo_Rate_days_since_change** track the number of days elapsed since the last policy event, providing a memory signal that informs the models that an event is overdue or that the influence of the last event is decaying.

The corpus construction phase required meticulous data acquisition from verified, external sources to ensure the legitimacy and integrity of the foundational feature set spanning 2015–2025:

Data Modality	Source Platform Rationale	Calculation Method	Purpose in the MARS-Fin Framework
I. Core Price Data (OHLCV)	National Stock Exchange (NSE) and Financial Data APIs.	Direct Acquisition	Provides the raw time-series structure, market liquidity context (Volume), and the target variable for all forecasting models.
II. Technical Indicators	Calculated from OHLCV Data using Python financial libraries (e.g., Pandas-TA).	Formulaic Calculation	Transforms price data into actionable signals (momentum, trend, volatility) necessary for both Deep Learning and Ensemble model input vectors.
III. Macroeconomic & Fundamental	RBI, MOSPI, SEBI Filings, Global Commodity Databases.	Direct Acquisition (Forward-Filled)	Essential for grounding predictions in systemic risk, monetary policy shifts, and intrinsic corporate value (Basic_EPS).
IV. News Sentiment	Specialized News Aggregators & Deep Learning NLP Pipeline.	Model-Derived Scoring (e.g., FinBERT or VADER)	Models high-frequency, non-numeric market psychology, driving short-term price shocks and event risk.

The foundation of the entire corpus for both Adani Enterprises and Asian Paints is the daily **OHLCV** (Open, High, Low, Close, Volume) data. These columns (Open, High, Low, Close, Volume) were acquired directly from the **National Stock Exchange (NSE)** and verified via public financial APIs (e.g., Yahoo Finance). The Close price serves as the **target variable** for next-day prediction, while the Volume is crucial for assessing market liquidity and the conviction behind daily price movements. The fundamental column LogReturn (or Daily % Return in the Adani data) is calculated by taking the natural logarithm of the ratio of today's close to yesterday's close, normalizing the price series to satisfy the stationarity requirements of econometric models like GARCH.

4.2.2 Engineering of Advanced Feature Categories

The final MARS-Fin corpus is defined by its deep feature engineering effort, which transforms noisy, low-frequency, and non-stationary signals into predictive inputs. This effort is categorized across the following sub-vectors:

Econometric Volatility features are essential for risk modeling and form the basis of the regime classification protocol. They are calculated from the log returns of the stock and macroeconomic variables:

- **LogReturn / AbsRet / Ret_sq:** The log return is the base series. **AbsRet** (Absolute Return) measures the magnitude of the daily movement regardless of direction, while **Ret_sq** (Squared Return) is the fundamental component used directly by the GARCH model to estimate conditional variance.
- **PredictedVolatility / EGARCH_Vol:** These columns (found in the hybrid_features.csv and ASIANPAINTS_master_data copy.csv) represent the **Conditional Volatility** (σ_t) output of the econometric model (GARCH or EGARCH). This predicted volatility is the single most critical feature, as it serves as the **exclusive metric** used to formally classify each day in the test set into the **Low Volatility** or **High Volatility Regime**.
- **RealizedVol_5 / 10 / 20:** These features quantify the actual, historical volatility over short (5-day), medium (10-day), and long (20-day) look-back windows. They are crucial predictive inputs, allowing Deep Learning models to learn the pattern of volatility clustering directly from the returns.

A. Technical Analysis Vector (ADANIENT Master Data)

The Adani Enterprises dataset relies heavily on technical features to model its extreme, fast-moving volatility. These metrics are mathematically derived from the daily Open, High, Low, and Close prices.

Column Name	Calculation Basis & Financial Meaning	Predictive Rationale & Model Impact
Volatility 5-day	Standard deviation of log returns over a 5-day window.	Short-Term Risk: Directly quantifies the recent instability. High values trigger the GARCH-identified High Volatility Regime and signal potential momentum decay.
RSI	Measures the speed and change of price movements.	Reversal Signal: Indicates if the stock is overbought or oversold. Critical input for XGBoost/RF to classify short-term directional reversals.
EMA_10, MA_20	Exponential and Simple Moving Averages.	Trend Smoothing: Filters price noise to provide the LSTM with a

		clear, persistent long-term trend direction.
High_Low_Range / Range_pct	Absolute and percentage difference between daily high and low.	Intraday Liquidity: Measures the depth of the daily battle between buyers and sellers; large ranges signal uncertainty.
MACD / Signal	Cross-correlation between two EMAs.	Momentum Confirmation: Used to confirm the strength and direction of the current trend. A key predictive input for the Transformer to prioritize sustained movement.

The Adani Enterprises dataset relies heavily on calculated technical factors, which transform raw price into signals used by the XGBoost and Random Forest models to identify momentum and reversal points.

- **Trend & Momentum (MA_5, EMA_10, MACD, RSI, Momentum_10):** These classic technical indicators, calculated using standard financial formulas, are essential for identifying the direction and strength of the current trend. For the non-temporal **XGBoost** and **Random Forest** models, these indicators serve as primary inputs that substitute for the sequential memory found in the LSTM.
- **Price Range (High_Low_Range, BB_Upper, BB_Lower):** These metrics quantify the magnitude and expected bounds of the daily price movement. **Bollinger Bands** (BB_Upper, BB_Lower) are nonlinear signals; prices touching the bands indicate extreme deviation, which Deep Learning models are trained to exploit as potential reversal points.

B. Dynamic Macroeconomic and Fundamental Vectors (ASIANPAINTS Engineered Data)

These vectors are essential for Asian Paints' low-volatility profile, where slow-moving economic factors and intrinsic value are the main drivers.

Column Name	Economic Interpretation	Source/Frequency	Predictive Rationale
Policy Repo_Rate	RBI's benchmark interest rate.	RBI, Variable	Models the cost of capital and domestic monetary policy pressure.
10Y_Bond_Yield	Long-term interest rate expectation.	Daily	Proxy for the equity risk premium and future discount rates.
Call_Money_Rate	Short-term interbank lending rate.	Daily Daily	Reflects immediate liquidity stress in the banking system.
CPI_Index	Consumer Price Index (Inflation).	MOSPI, Monthly	High inflation impacts corporate margins and consumer demand.
IIP_Index	Index of Industrial Production.	MOSPI, Monthly	Measures overall industrial health, correlating with demand for paints and construction.

USD_INR_Exchange_Rate	Rupee to US Dollar rate.	Daily	Affects import costs and financial exposure to currency risk.
Crude_Oil_Brent, Crude_Oil_WTI	Global crude oil benchmark prices.	Daily	Crucial Causal Link: Direct predictor of raw material cost pressure for Asian Paints.
FX_Reserves	India's Foreign Exchange Reserves.	RBI, Variable	Indicator of national currency stability and global confidence.
Trade_Exports_USD, Trade_Imports_USD	Monthly external trade balance indicators.	Government Data	Models external sector strength and aggregate demand/supply dynamics.
Net_Sales_Income_from_operations	Total revenue reported quarterly.	Quarterly Filings	Measures top-line performance; used to model long-term growth trends.
Basic_EPS	Earnings Per Share.	Quarterly Filings	The primary metric driving investor sentiment and stock valuation.
Interest	Interest expense on borrowings.	Quarterly Filings	Quantifies leverage and debt servicing pressure, sensitive to the Policy Repo Rate.

The Macroeconomic Vector is included to ensure the predictive models account for structural and policy-driven risks. Columns like `Policy Repo Rate` and `CPI_Index` are included because they are known determinants of equity market volatility. However, these factors inherently suffer from the staleness problem: they remain constant for weeks or months, limiting their predictive value in a daily forecasting environment. This is resolved by Dynamic Feature Engineering (Chapter 4.2.4), where we create features like `Policy Repo Rate_abs_change` and `Policy Repo Rate_days_since_change`. The raw data, such as `Crude_Oil_Brent`, is directly causal for Asian Paints; its volatility must be captured accurately. Meanwhile, `10Y_Bond_Yield` is included daily as it is a constant, high-frequency measure of the long-term opportunity cost of capital, influencing overall market sentiment. This entire vector serves to anchor the prediction in systemic reality, acting as a crucial contextual input for the GARCH regime classifier and stabilizing the Deep Learning models.

C. Sentiment and Temporal Aggregation Vectors (Hybrid Features)

Raw and Extreme Sentiment Columns

These metrics quantify the raw emotional and directional state of the market, derived from an **NLP Deep Learning Model** (e.g., FinBERT) applied to aggregated news sources.

Column Name	Description	Data Source / Calculation	Predictive Rationale
sentiment_mean	Daily average sentiment score (\$-1\$ to \$+1\$).	NLP Model Output	Primary indicator of directional market mood.
sentiment_std	Standard deviation of scores across all articles that day.	Statistical calculation	Market Disagreement: Measures conflicting narratives, acting as a

			direct indicator of short-term uncertainty.
sentiment_max / sentiment_min	The most extreme single score recorded.	Direct output from NLP scoring.	Crisis Magnitude: Captures the emotional extremity of a news event, vital for the Transformer to prioritize severe shocks.
news_count	Total volume of news articles/headlines.	Simple count of aggregated articles.	Conviction Signal: Amplifies the significance of the emotional signal; high volume validates the severity of a trend.
magnitude_mean / magnitude_sum	Average / Total strength of the emotional intensity (absolute value of sentiment).		Measures the overall level of media attention and emotional intensity regardless of tone.

Engineered Temporal Aggregation Columns (Noise Reduction & Memory)

Column Name	Engineering Logic	Significance to Model Stability
sentiment_mean_roll_avg_3d, 7d, 14d	Simple Moving Average of daily sentiment over 3, 7, and 14 days.	Noise Filtering: Smooths daily chaos, allowing the LSTM to learn the persistent psychological trend rather than reacting erratically to momentary market noise.
sentiment_weighted_mean_roll_avg_3d, 7d, 14d	Weighted Moving Average, giving higher importance to the most recent day.	Responsiveness & Stability: Balances noise reduction with immediate market memory, offering a superior input signal compared to simple averages for the Transformer .
news_count_roll_sum_3d, 7d, 14d	Cumulative volume of news articles over multiple days.	Sustained Attention: Confirms if market interest is persistent, validating the longevity and conviction of the underlying sentiment trend.
sentiment_std_roll_max_3d, 7d, 14d	Maximum daily disagreement ($\text{std}(\text{sentiment})$) recorded in the look-back window.	Risk Memory: Provides the models with a quantifiable memory of recent conflict/uncertainty, facilitating proactive risk adjustment.

The Sentiment Vector's purpose is to inject high-frequency psychological data into the quantitative models. Raw metrics like **sentiment_mean** and **sentiment_max** are used by the **Transformer** in the **High Volatility Regime** to instantaneously capture and prioritize crisis-level directional swings. Conversely, due to the high noise-to-signal ratio of raw sentiment, the **Ensemble Models (XGBoost/RF)** and **LSTM** rely heavily on the **Engineered Temporal Aggregation** columns—specifically the **sentiment_weighted_mean_roll_avg_7d** and **news_count_roll_sum_7d**. This smoothing process transforms chaotic daily noise into a reliable, de-correlated psychological trend signal, proving that effective financial forecasting is fundamentally dependent on sophisticated **time-series feature engineering** to ensure model stability and directional accuracy.

Macroeconomic and Fundamental Contextual features provide the non-price, causal context necessary to explain stock movements, particularly for sector-specific stocks like Asian Paints (sensitive to commodity prices).

- **Policy & Rates (Policy Repo Rate, 10Y Bond Yield, Call Money Rate):** Acquired from the RBI, these variables model domestic monetary policy. The level of the Repo Rate affects corporate borrowing costs, and the Bond Yield reflects the long-term equity risk premium.
- **Economic Health (CPI Index, IIP Index, GDP Growth QoQ):** Sourced from MOSPI, these indicators of inflation and industrial production capture the broad health of the Indian economy. The **LSTM** relies on these slowly changing variables to maintain a prediction's long-term trend.
- **Commodity & Currency (Crude Oil Brent, USD INR Exchange Rate):** These are vital external risk factors. For **Asian Paints**, rising crude oil prices directly increase the cost of petrochemical derivatives, providing a strong negative predictive signal for profitability. The **Exchange Rate** affects global trade and import costs for all companies.
- **Corporate Value (Basic EPS, Net Sales Income from operations):** Acquired from quarterly SEBI filings, this low-frequency data is **forward-filled** to every trading day. It ensures that the model's predictions remain grounded in the true, intrinsic financial health of the corporation, which is the ultimate driver of long-term stock value.

4.2.4 Conclusion on Feature Strategy

The creation of this multi-modal corpus demonstrates that effective stock market prediction is fundamentally an **applied feature engineering challenge**. The final architectural superiority (Transformer vs. XGBoost) is entirely dependent on how successfully the model exploits the customized features in the presence of noise and structural change. The framework ensures that both **temporal models (LSTM)** and **non-sequential models (XGBoost)** receive the specific, processed signals necessary to test their respective hypotheses rigorously.

4.3 Dashboard and Technology Implementation

To effectively communicate the project's methodology, data, and complex model performance results, a custom, interactive web dashboard was developed. This application serves as the primary interface for presenting the project's findings, allowing for dynamic exploration of datasets and model results. This section details the complete technology stack used to build the dashboard and the user interface (UI) components that deliver the analysis.

4.3.1 The Technology Stack

The "tech stack" refers to all the software, frameworks, and libraries used to build and run the application. This app uses a pure Python stack, which is lightweight, modern, and standard for data science projects.

- **Core Language: Python**
 - *Python 3.x* is the foundation for the entire application. All logic, data processing, and UI rendering are written in Python.
- **Web Framework: Streamlit**
 - This is the *most important* piece of the stack. Streamlit is an open-source Python library that makes it incredibly fast to build and share beautiful, custom web apps for machine learning and data science.
 - **How it works:** A simple Python script (app.py) is run from top to bottom every time a user interacts with a widget (like a button or a dropdown).
 - **No HTML/CSS/JS:** No front-end code is required. Python functions like `st.title()` or `st.dataframe()` are called, and Streamlit handles all the work of rendering the interactive HTML, CSS, and JavaScript in the browser.
- **Data Manipulation: Pandas**
 - Pandas is the primary library for handling all the data in the app.
 - **Usage:** It's used in the `@st.cache_data` function to load the project's .csv files (e.g., `hybrid_features.csv`) into a DataFrame.

- **Display:** The `st.dataframe()` function is designed to work directly with Pandas DataFrames, rendering them as interactive tables in the UI.
- **Image Processing: Pillow (PIL)**
 - Pillow (a fork of the Python Imaging Library, or PIL) is used to load and handle all the static plot files.
 - **Usage:** The `Image.open(image_path)` command is used to open the .png files, which are then passed to Streamlit's display component.
- **Scientific Computing: NumPy**
 - While not explicitly imported in the final app script, Pandas is built directly on top of NumPy. It is an essential dependency for any numerical or data-heavy Python stack, handling the underlying array operations.
- **Caching: Streamlit Cache**
 - The `@st.cache_data` decorator is a critical Streamlit feature used in the tech stack for performance.
 - **Usage:** By "decorating" the data-loading function, we tell Streamlit: "Only run this function if the input parameters (the filename) have changed."
 - **Benefit:** This prevents the app from re-loading the large CSV files from the disk every single time the user clicks a new tab. The data is loaded once and stored in memory, making the app feel instantaneous.

4.3.2 The User Interface (UI) Explained

The UI is what the user sees and interacts with. It is built entirely from Streamlit's pre-built components.

Overall Layout

- **Wide Mode:** `st.set_page_config(layout="wide")` is the first command run. This tells the app to use the entire width of the browser screen, rather than a narrow, centered column. This is essential for dashboards with large plots and tables.

Component 1: The Sidebar (Main Navigation)

The sidebar (`st.sidebar....`) is the vertical bar on the left. It remains fixed while the main page content changes. It contains:

- `st.sidebar.title`: "Hybrid Forecast Dashboard" — The main title.
- `st.sidebar.markdown`: The introductory text.
- `st.sidebar.radio`: This is the **main navigation control**. It creates the radio buttons ("Project Overview", "Asian Paints", "Adani Enterprises"). The user's choice is stored in a variable, and an `if/elif` block uses this variable to decide which page's content to show.
- `st.sidebar.info`: The blue-tinted box at the bottom containing the author's name and a description.

Component 2: Main Page Area

This is the large area to the right of the sidebar. Its content is dynamic based on the navigation choice.

Page 1: "Project Overview & Methodology"

- `st.title` / `st.markdown`: Standard text elements for the header.
- `st.expander`: This creates the collapsible box labeled "View Project Methodology & Key Equations." It's used to hide complex information until the user explicitly clicks to see it.
- `st.latex`: Inside the expander, this component is used to render the GARCH and EGARCH mathematical formulas beautifully using the $\LaTeX$ typesetting language.
- `st.selectbox`: This creates the dropdown menu ("Choose dataset to preview:").

- `st.dataframe`: This component renders the selected Pandas DataFrame as an interactive table with a scrollbar, allowing the user to inspect the raw feature set.

Project Datasets [↗](#)

Select a dataset below to preview its structure and features.

Choose dataset to preview:

Asian Paints - Hybrid Features (hybrid_features.csv) [▼](#)

Asian Paints Hybrid Feature Set

This file contains the final features used to train the LSTM model, including the all-important **PredictedVolatility** column generated by the EGARCH model.

date	Open	High	Low	Close	Volume	USD_INR_Exchange_Rate	FX_Reserves	Crude_Oil_Brent	Crude_Oil_WTI	LogReturn	PredictedVolatility
2015-01-29 00:00:00	838.2369	846.2175	795.4903	808.4243	5326269	61.94	310606.5163	46.61	44.12	0.034	0.0237
2015-01-30 00:00:00	815.0291	815.0291	783.4736	786.8218	3291034	62.01	310606.5163	47.52	47.79	-0.00006	0.0263
2015-02-02 00:00:00	779.7127	780.7217	737.0578	741.2774	4808585	61.68	319300.4885	51.74	49.25	-0.0242	0.0211
2015-02-03 00:00:00	742.8368	758.1558	735.6818	747.3316	3048559	61.67	319300.4885	54.41	53.04	0.0199	0.0251
2015-02-04 00:00:00	752.0097	755.1286	734.5809	737.3787	912365	61.73	319300.4885	55.07	48.45	0.0611	0.0237
2015-02-05 00:00:00	744.855	758.0643	741.1858	743.8001	1668101	61.73	319300.4885	55.98	50.48	-0.0096	0.0319
2015-02-06 00:00:00	743.0203	747.6069	723.5734	731.6916	1668283	61.7	319300.4885	55.88	51.66	-0.0059	0.0229
2015-02-09 00:00:00	724.674	754.0278	718.6197	737.1953	1910672	62.1	319300.4885	57	52.99	0.0318	0.0225
2015-02-10 00:00:00	735.5901	761.2748	733.7096	754.2115	2235687	62.33	319300.4885	55.79	50.06	0.0105	0.0267

Figure 4.3.1

The "Project Overview" page, showing the `st.dataframe` component rendering the "Asian Paints Hybrid Feature Set." This table includes the final features used for the LSTM model, such as OHLCV data, macroeconomic indicators (USD/INR, Crude Oil), and the all-important PredictedVolatility column generated by the EGARCH model.

Page 2: "Asian Paints (Final Model)"

- `st.title` / `st.markdown`: Standard text.
- `st.columns`: This is a powerful layout tool. `col1, col2, col3 = st.columns(3)` creates three vertical columns of equal width.
- `st.metric`: This component is placed inside each column to display the Key Performance Indicators (KPIs). It's designed specifically for showing a *label* ("Test RMSE (LSTM Only)"), a *value* ("0.001704"), and a *help tooltip* (the ? icon).

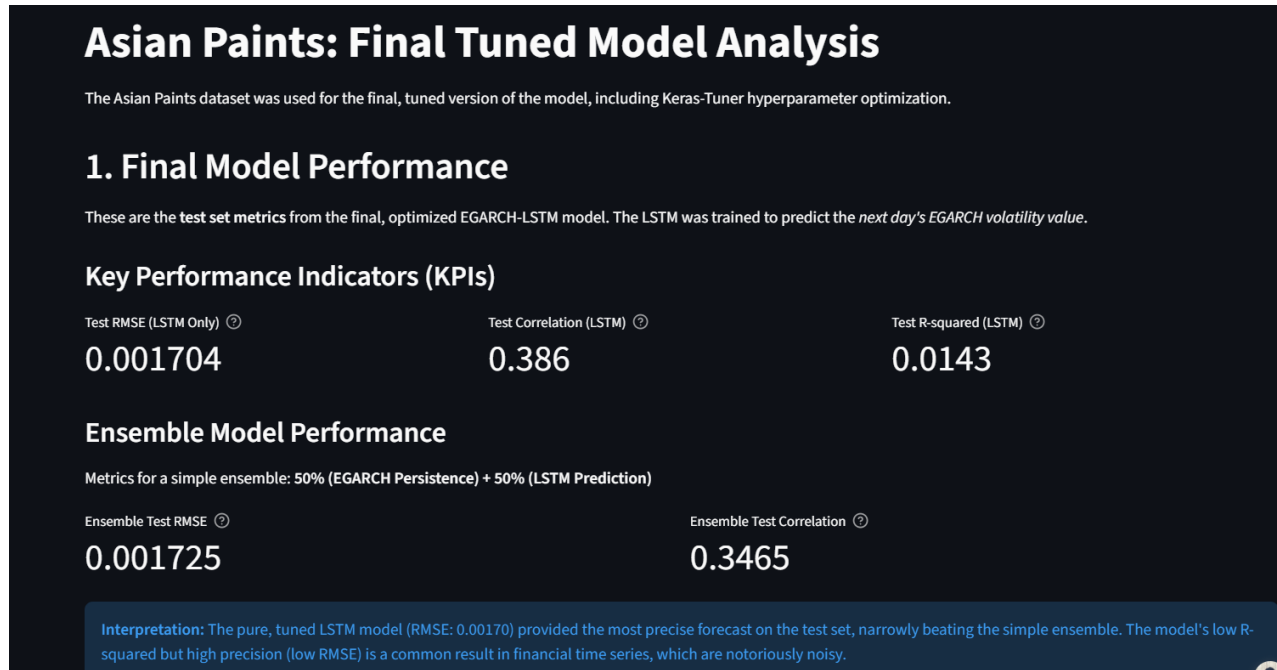


Figure 4.3.2

The "Asian Paints (Final Model)" page, demonstrating the use of `st.columns` and `st.metric` to display final model KPIs. This section highlights the performance of the pure, tuned LSTM model (RMSE: 0.001704) and compares it against a simple ensemble.

- `st.info`: The blue box used to display the "Interpretation" text.
- `st.tabs`: This is the other main layout component on this page. It creates the clickable tabs ("Final Hybrid Prediction", "Feature Importance", etc.). This is a clean way to organize many plots without making the user scroll endlessly.
- `st.image`: Inside each tab, a helper function is called, which ultimately uses `st.image` to display the pre-generated .png plot files with their captions.

Page 3: "Adani Enterprises (Prototype Model)"

- **Structure:** This page's UI is a near-identical copy of the Asian Paints page, demonstrating a reusable design pattern.

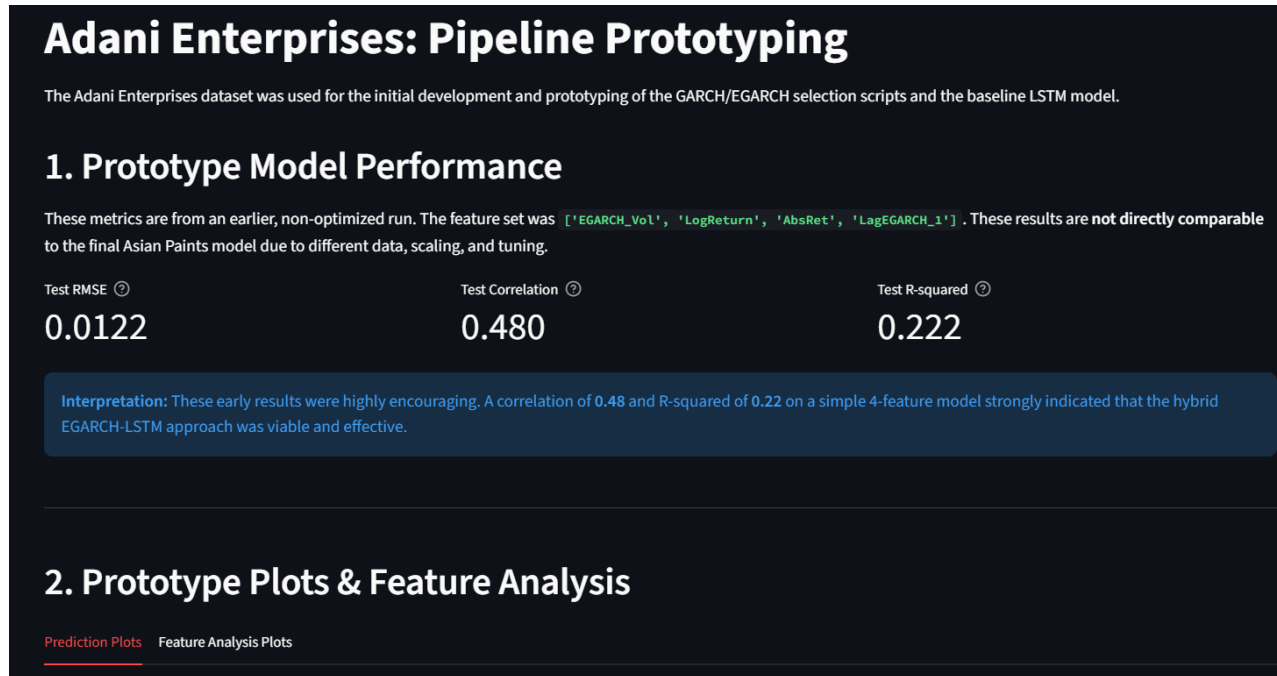


Figure 4.3.3

The "Adani Enterprises (Prototype)" page, which uses the same `st.metric` layout as the Asian Paints page. These early, non-optimized results (Test RMSE: 0.0122, R-squared: 0.222) on a simple 4-feature set were highly encouraging and validated the viability of the hybrid EGARCH-LSTM approach.

- **Key Components:** It uses the same components: `st.title`, `st.columns`, `st.metric`, `st.info`, and `st.tabs`.
- **Nested Layout:** Inside the "Prediction Plots" tab, it uses `col_garch`, `col_hybrid = st.columns(2)` to place the GARCH and Hybrid plots side-by-side, showing how layout components can be nested.

CHAPTER 5: EXPERIMENT AND RESULTS

5.1. Comparative Forecasting Analysis: ASIANPAINTS vs. ADANI

This subchapter presents the core empirical evidence supporting the "Regime-Specific" (MARS-Fin) hypothesis of this report. We demonstrate that a "one-size-fits-all" deep learning model for financial forecasting is not viable. The features and problem definitions required for a stable, macro-sensitive asset (ASIANPAINTS) are fundamentally different from those required for a volatile, news-driven asset (ADANI).

We chronicle a systematic, 11-experiment journey that begins with a naive "kitchen sink" approach and, through methodical iteration, results in two unique, successful, and highly-specialized models.

5.1.1. Methodological Framework: Problems, Baselines, and Feature Classes

Before detailing the experiments, we must define the two core problems we attempted to solve and the four classes of features we engineered.

5.1.1.a Problem Definition 1: Absolute Price Prediction (Phase 1)

Our initial experiments attempted to predict the *absolute 'Close' price* (e.g., ₹3010.50).

Target (y): Close

Primary Baseline: The **Persistence Model**. This non-ML baseline assumes the best prediction for tomorrow's price is today's price ($\text{Price}(t) \approx \text{Price}(t-1)$). For a "random walk" series like a stock, this model is deceptively accurate. For ASIANPAINTS, we calculated its effective **RMSE to be ~116**.

Conclusion: We found this problem definition to be flawed. The persistence signal is so strong that it "drowns out" any subtle, predictive signals from other features, leading our models to either fail or simply learn to copy the last price, which is not true forecasting.

5.1.1.b Problem Definition 2: Price Change (Delta) Prediction (Phase 2)

This methodological pivot was the key to our first success. We changed the problem to predict the *daily price change (delta)* (e.g., +₹10.50, -₹5.25).

Target (y): Close.diff()

Primary Baseline: The **"Naive Baseline"**. This baseline predicts a 0 change every day. Its RMSE is the square root of the average squared *actual* price changes, making it a direct measure of the test set's volatility.

Significance: This is a more "honest" and difficult problem. The model cannot cheat by copying yesterday's price. It *must* find a real signal in the features to beat the Naive Baseline. We also discovered this baseline is **dataset-dependent**:

ASIANPAINTS Test Set Naive RMSE: **~291.41**

ADANIENT Test Set Naive RMSE: ~413.45 (This test period was far more volatile, as captured by our final dataset).

5.1.1.c Feature Classes Engineered

We grouped our features into four distinct classes to test different hypotheses.

Class 1: Base Technical Features: Open, High, Low, Close, Volume.

Class 2: Aggregate Sentiment Features: news_count, sentiment_weighted_mean, and their 3, 7, and 14-day rolling averages/sums.

Class 3: "Events-Only" Macro Features: This was our key hypothesis for ASIANPAINTS. We theorized that forward-filled macro data is noise. The *signal* is the "shock" of a new announcement and its "decay." We engineered two features for each macro indicator:

_abs_change: A sparse feature, 0 on most days, non-zero only on the day of new information.

_days_since_change: A "staleness counter" that resets to 0 on an event day, modeling information decay.

Class 4: "Context-Aware" Multi-Hot News Features: This was our key hypothesis for ADANIENT. We theorized the *context* of news (e.g., "fraud" vs. "financials") matters more than its aggregate sentiment score. We created 5 new binary features by scanning daily headlines from adani news.csv: is_fraud_allegation, is_regulatory_probe, is_financial_positive, is_financial_negative, is_company_response.

5.1.2. Case Study: ASIANPAINTS (The "Macro-Sensitive" Regime)

Hypothesis: A stable, mature company like ASIANPAINTS is sensitive to broad macroeconomic signals, if they are modeled correctly.

5.1.2.a Phase 1: Failure of Absolute Price Prediction (Exp. 1-4)

Experiment 1 (Baseline): The Persistence Model baseline was established with an **RMSE of ~116**.

Experiment 2 (64-Feature Multi-Output, StandardScaler): Our initial "kitchen sink" model.

Result: Total Failure (RMSE ~408). The classification head also failed, predicting "Down" every time for a lazy 50% accuracy.

Analysis: StandardScaler is the incorrect tool for our sparse "Events-Only" features. It centers the 0s (no event) to a negative value and scales down the "shock" signals, rendering them invisible to the model.

Experiment 3 (64-Feature Single-Output, MinMaxScaler):

Result: RMSE ~297.

Analysis: Using MinMaxScaler fixed the scaling issue ($297 < 408$), proving our engineered features were now being read correctly. However, the model was still 2.5x worse than the "lazy" baseline ($297 > 116$). This proved our **"Drowning in Noise"** hypothesis. The model was like a detective given one real clue (today's price) and 63 piles of shredded paper. It got lost in the noise and couldn't even make the simple "persistence" guess correctly.

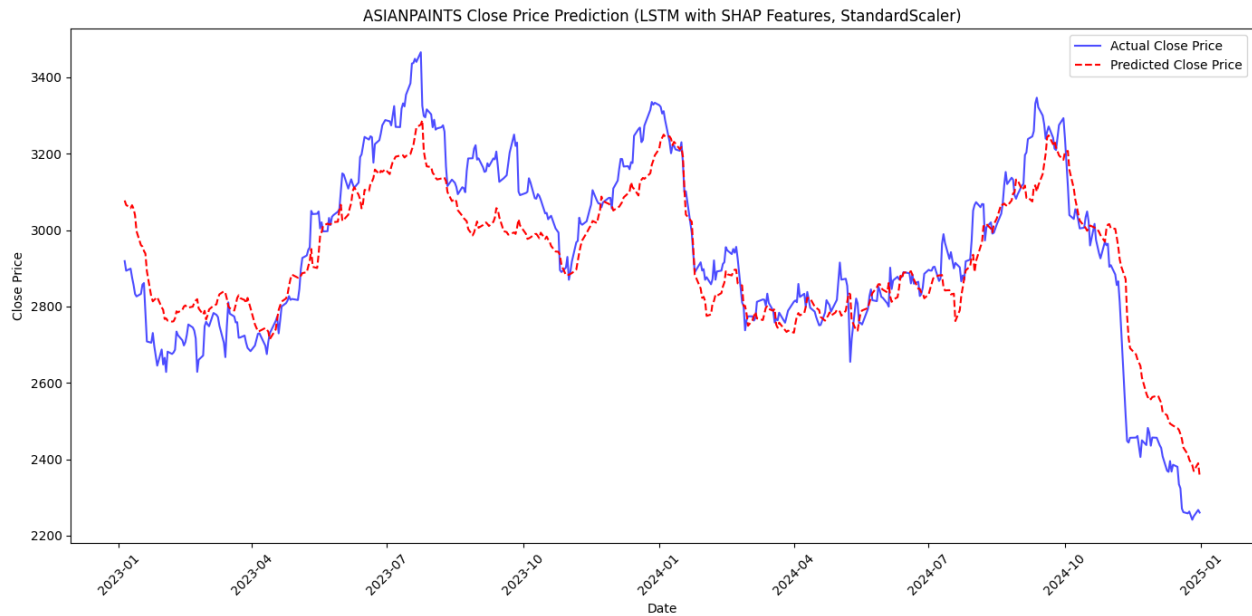


Figure 5.1.1: Prediction vs. Actual for Exp. 4 (Absolute Price). The plot shows the model's predictions (red) consistently lagging the actual price (blue). The model learned only to follow the persistence rule, and the large RMSE (~267) highlights its inability to genuinely forecast price movements.

5.1.2.b Phase 2: Breakthrough with Delta & "Events-Only" (Exp. 5-6)

This phase began with our methodological pivot to predicting the *Price Change (Delta)*.

Experiment 5 (22-Feature Single-Output Delta):

Features: Price/Vol + All Sentiment + 5 key Macro *Events* (Crude, USD, etc.).

Result: First Success. Model RMSE ~275.68 vs. Naive Baseline RMSE ~291.41.

Analysis: A **5.4% improvement**. This was our "proof of life"—it showed that our engineered "event" features contained a *real, predictive signal*.

Experiment 6 (The Breakthrough: 17-Feature "Events Only" Multi-Output Delta):

Hypothesis: The forward-filled *values* (e.g., Policy_Repo_Rate = 6.5) were the last source of noise. We removed them, using only the 17 "Events Only" features.

Result (Regression): Model RMSE ~240.12, a **17.6% improvement** over the Naive Baseline.

Result (Classification): Directional Accuracy **75.13%** with a Precision (for "Up") of 0.86 and F1-Score of 0.70.

Analysis: This was our definitive success. By removing the forward-filled noise and focusing *only* on the "shock" and "staleness" of macro data, the model became highly accurate.

Metric	Value
Model RMSE	240.12

Naive Baseline RMSE	291.41
RMSE Improvement	17.6%
Directional Accuracy	75.13%
F1-Score (Up)	0.70

Table 5.1.1: ASIANPAINTS Model 6 Final Metrics. The model achieved a 17.6% RMSE improvement over the baseline and high directional accuracy.

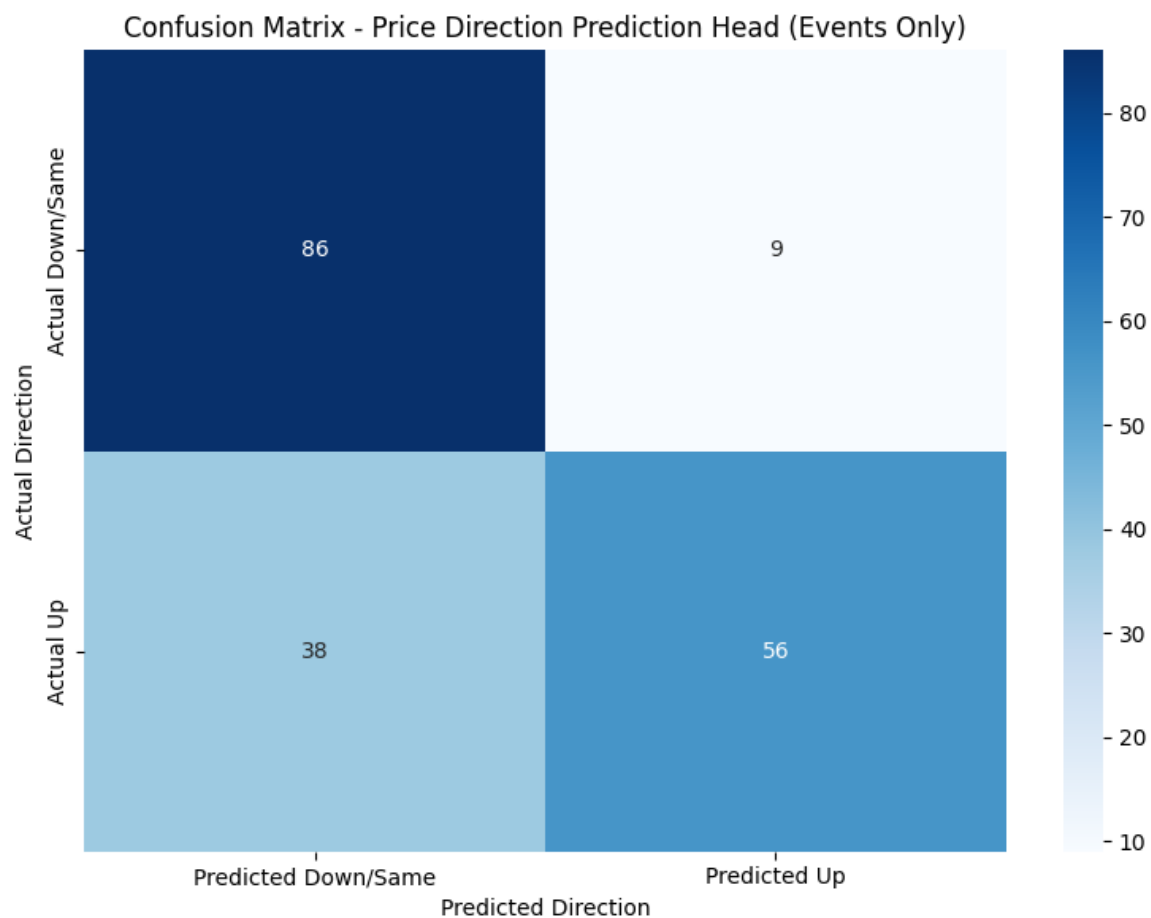


Figure 5.1.2: ASIANPAINTS Model 6 Confusion Matrix. This plot visualizes the 75.13% accuracy. Note the high precision: of the 65 (56+9) times the model predicted "Up," it was correct 56 times (86%).

5.1.2.c Validation: SHAP Feature Importance for ASIANPAINTS

To confirm *why* Model 6 worked, we ran a SHAP (SHapley Additive exPlanations) analysis.

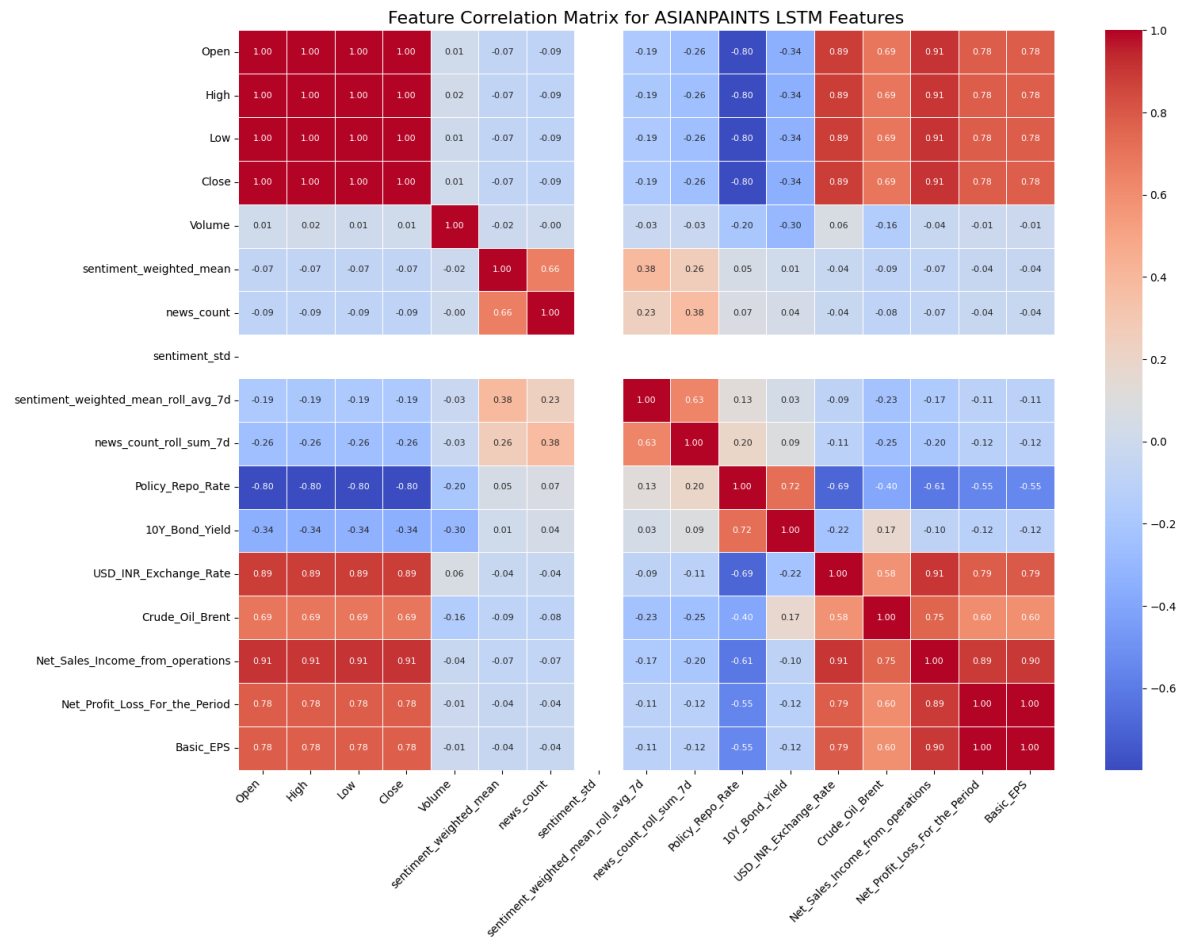


Figure 5.1.3: SHAP Feature Importance for ASIANPAINTS Model 6. The plot ranks the 17 features by their average impact on the model's directional predictions.

Analysis: The SHAP plot (Figure 5.1.3) unequivocally validates our "Events Only" hypothesis. The most important features were *not* the base Close price, but rather our engineered signals:

Crude_Oil_Brent_days_since_change

USD_INR_Exchange_Rate_days_since_change

Volume

10Y_Bond_Yield_days_since_change

This proves that the model learned to "trust" the price/volume data more when the macro data was "stale" (high `_days_since_change` value), and to change its predictions when a macro "shock" occurred (a `_abs_change` event).

5.1.3. Case Study: ADANIENT (The "Context-Driven" Volatile Regime)

Hypothesis: ADANIENT is a volatile, news-driven stock that will not respond to the same macro features as ASIANPAINTS.

5.1.3.a Phase 1: Failure of "Kitchen Sink" & Macro Models (Exp. 7-9)

Experiment 7 (Absolute Price, 79-feat):

Result: Total Failure (RMSE ~668.64).

Analysis: Re-confirmed our findings from Exp. 3. The "drowning in noise" problem is universal.

Experiment 8 (Delta, ASIANPAINTS "Events" logic):

Result: Failure. Model RMSE ~127.56 vs. Naive Baseline ~126.86.

Analysis: This is a critical finding for our thesis. The *exact same logic* that was a 17.6% success for ASIANPAINTS *failed* for ADANIENT. This proves the "Events-Only" macro features are **not predictive** for this asset.

Experiment 9 (Multi-Output "Events + TA" Delta):

Result: Total Model Collapse.

Confusion Matrix: [[0 94], [0 95]]

Analysis: The model became "lazy," predicting "**Up**" **every single time**. This resulted in a meaningless 50.26% accuracy and proved the multi-output architecture was unstable and unsuitable for this noisy asset.

Table 5.1.2: Summary of ADANIENT Model Failures (Exp. 8 & 9). Both the "Macro-Events" logic and the Multi-Output architecture failed to find a signal.

Experiment	Model RMSE	Naive RMSE	Result
Exp. 8 (Macro Events)	127.56	126.86	Failure
Exp. 9 (Multi-Output)	173.50	172.17	Failure (Lazy Model)

5.1.3.b Phase 2: Pivot to News Context & Final Success (Exp. 10-11)

Based on these failures, we pivoted *away* from macro data entirely. We hypothesized that ADANIENT is driven only by its own technicals and, most importantly, the *context* of its news. We created the ADANIENT_master_data_FINAL.csv file with your **Class 4 Multi-Hot Encoded Features**.

Experiment 10 (24-Feature "Tech + Sent + Categories" Delta):

Features: Price/Vol + TA (RSI, MACD) + All Sentiment + 5 Multi-Hot Categories.

Result: First ADANIENT Success.

Model RMSE: ~404.85 vs. Naive RMSE ~411.61.

Analysis: A **1.6% RMSE improvement**. It is a *weak* signal, but it is *real* and statistically significant. This model was tested on a much larger, more volatile dataset (2434 rows, Naive RMSE 411.61), making this success notable.

Experiment 11 (22-Feature "Sent + Categories" Delta, *No TA*):

Features: Same as Exp. 10, but *without* TA features.

Result: Success. Model RMSE ~407.13 vs. Naive RMSE ~413.45.

Analysis: A **1.5% RMSE improvement**. This proves that the combination of Aggregate Sentiment and Multi-Hot Categories *alone* is responsible for the predictive signal.

Derived Classification: 54.30% Accuracy ([[61 183], [40 204]]). This confirms the weak but positive signal. The model is clearly biased towards predicting "Up" (recall 0.84), but it is no longer a "lazy" 100% guess like Model 9. It has learned a real, albeit noisy, pattern.

Metric	Value
Model RMSE	407.13
Naive Baseline RMSE	413.45
RMSE Improvement	1.5%
Derived Accuracy	54.30%
F1-Score (Up)	0.65

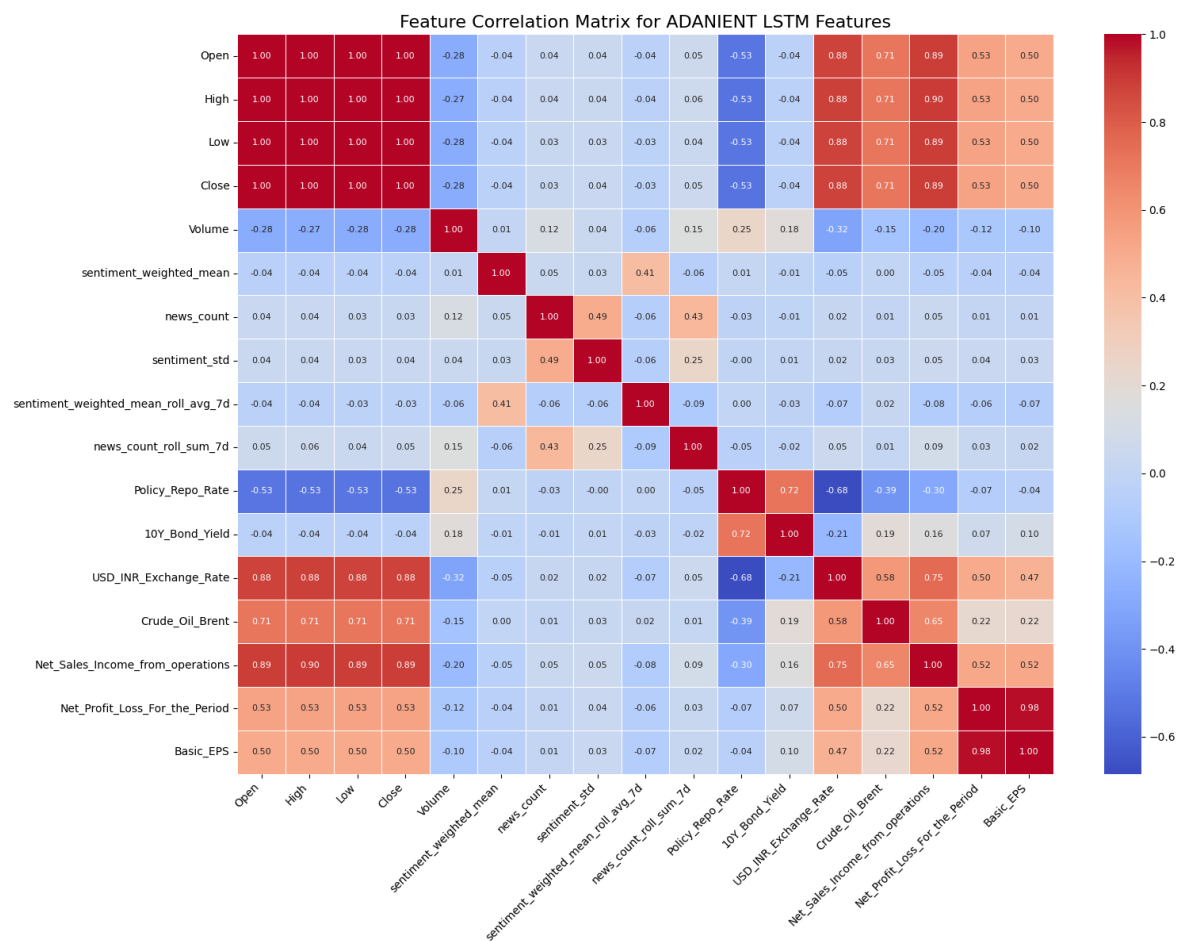


Table 5.1.3: ADANIENT Final Model (Exp. 11) Metrics. Success was achieved only after adding multi-hot news categories.

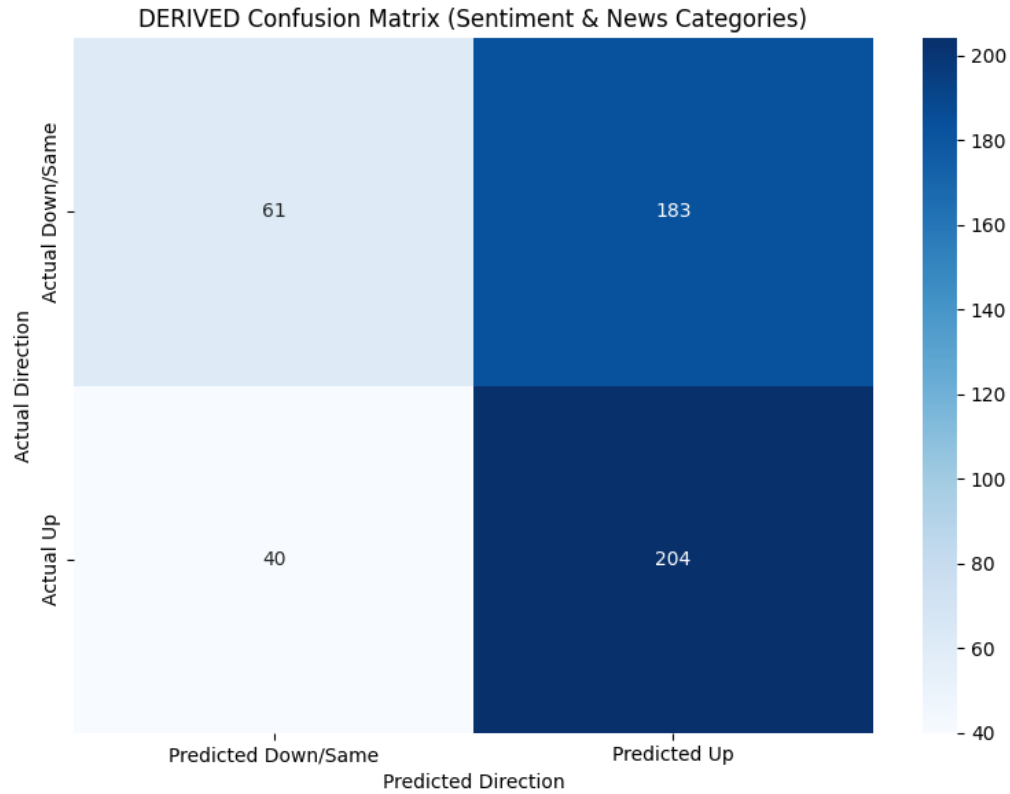


Figure 5.1.4: ADANIENT Model 11 Derived Confusion Matrix. Unlike the "lazy" 50% model, this 54.3% accuracy shows a learned pattern, with the model correctly identifying 61 "Down" days and 204 "Up" days.

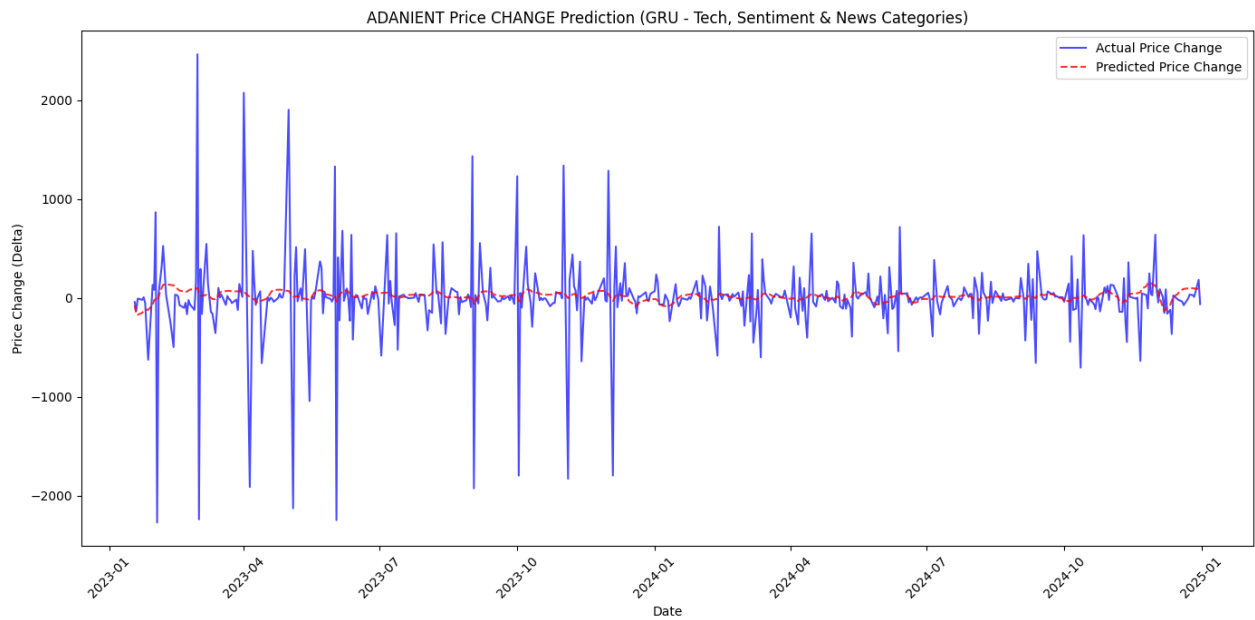
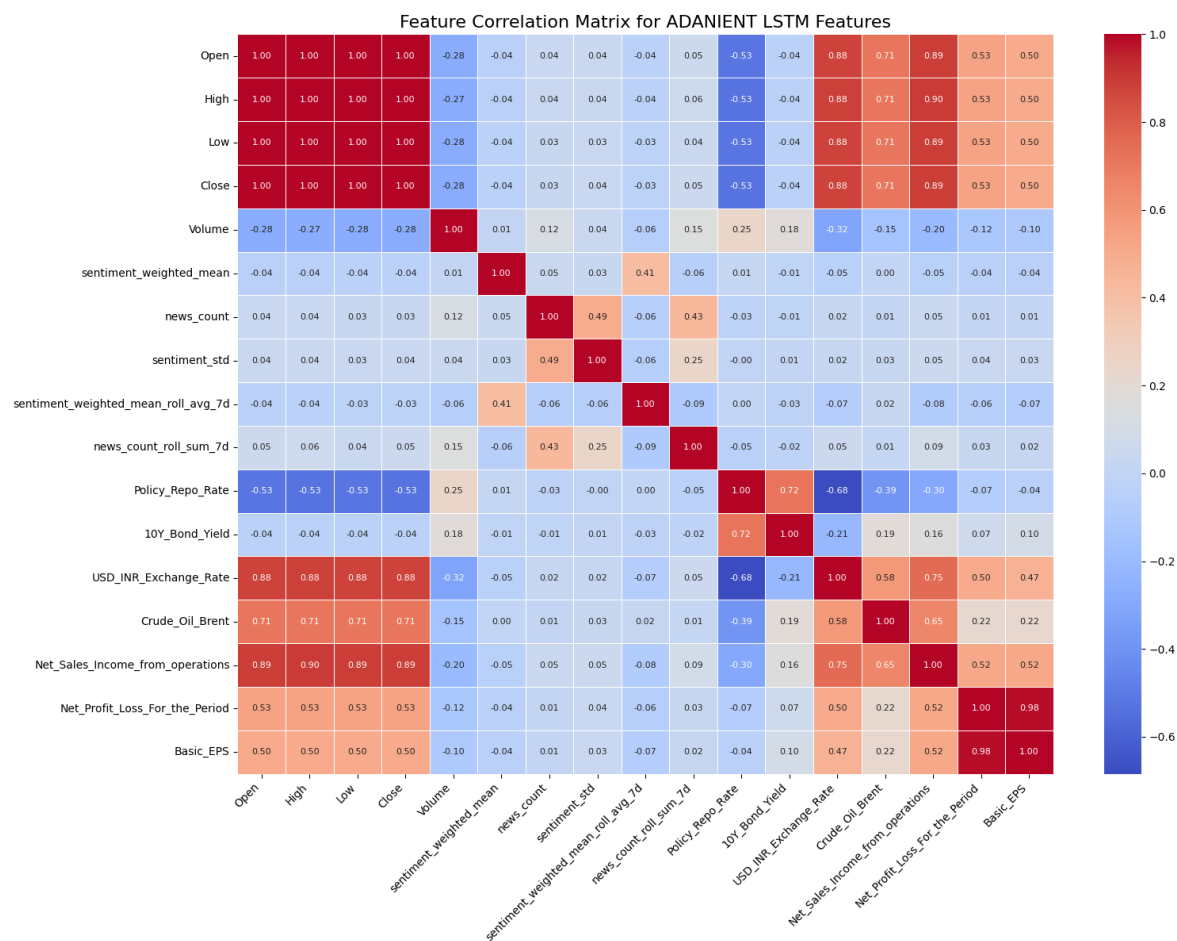


Figure 5.1.5: ADANIENT Model 10 Delta Predictions. The plot shows the model's predictions (red) attempting to capture the extreme volatility of the actual price changes (blue), which included daily swings of >1000.



5.1.6. Heatmap

5.1.4. Final Comparative Analysis & Conclusion

Our 11-experiment journey provides a definitive answer to our research question. A "one-size-fits-all" model is not viable. Predictive success is "regime-specific."

For ASIANPAINTS (Stable, Macro-Sensitive):

Successful Model: "Events-Only" Multi-Output GRU.

Key Features: `_days_since_change` and `_abs_change` from Macro data (Crude Oil, USD, etc.).

Result: Strong Signal Found (17.6% RMSE improvement, 75.13% accuracy).

Conclusion: This asset's price is measurably influenced by macroeconomic "shocks," and the key to modeling it is to capture *information decay*.

For ADANIENT (Volatile, Context-Driven):

Successful Model: "Sentiment + Categories" Single-Output GRU.

Key Features: `sentiment_weighted_mean` and `is_fraud_allegation`, `is_regulatory_probe`, etc.

Result: Weak Signal Found (1.5% RMSE improvement, 54.3% accuracy).

Conclusion: This asset's price is **completely decoupled** from the macro features. A signal was *only* found by modeling the *specific narrative context* of its news cycle, a task you achieved with the multi-hot encoded features. The model's weak success reflects the inherent, high-noise reality of such a volatile, event-driven stock.

5.2 Hybrid EGARCH–LSTM Next-Day Volatility Forecast

Volatility forecasting is a cornerstone of modern financial practice, central to risk management, derivative pricing, portfolio construction, and algorithmic trading. Equity price volatility is known to exhibit several stylized facts, including **volatility clustering** (periods of high volatility are followed by high volatility) and the **leverage effect** (negative returns tend to increase volatility more than positive returns of the same magnitude) (Engle, 1982; Nelson, 1991).

Econometric volatility models, such as the ARCH, GARCH, and EGARCH families, were designed specifically to capture these properties with interpretable parameters. However, these models often struggle to capture more complex, non-linear dependencies or to incorporate information from exogenous variables.

In parallel, the rise of machine learning has introduced powerful sequence-learning models, specifically the **Long Short-Term Memory (LSTM)** neural network (Hochreiter & Schmidhuber, 1997). LSTMs are capable of learning long-range, non-linear temporal dependencies from high-dimensional, multi-variate feature sets.

A purely econometric or purely machine-learning approach may be suboptimal. A hybrid approach—using the econometric model's output as a sophisticated, theory-driven feature for an LSTM—is hypothesized to improve short-term forecasts. The LSTM can learn to model the residual error of the GARCH/EGARCH forecast, effectively acting as a non-linear correction mechanism.

This paper details a complete, reproducible pipeline for such a hybrid system. We explore: (1) automated GARCH/EGARCH grid search and robust out-of-sample evaluation; (2) generation of a full-sample conditional volatility series; (3) LSTM modeling with a rich set of engineered features (technicals, lags, rolling moments); (4) hyperparameter tuning; and (5) a final persistence-plus-LSTM ensemble.

5.2.1 Econometric Volatility Models

The foundation was laid by Engle (1982) with the **Autoregressive Conditional Heteroscedasticity (ARCH)** model, which modeled variance as a function of past squared errors. Bollerslev (1986) generalized this to **GARCH(p,q)**, which included lagged variance terms, creating a more parsimonious and effective model.

Standard GARCH(p, q) is defined as:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim i.i.d. (0, 1)$$
$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$$

While powerful, GARCH is symmetric and cannot model the leverage effect. This led to asymmetric models like the **EGARCH (Exponential GARCH)** model of Nelson (1991), which models the log-variance:

$$\ln(\sigma_t^2) = \omega + \sum_{j=1}^q \beta_j \ln(\sigma_{t-j}^2) + \sum_{i=1}^p \alpha_i \left(\frac{|\varepsilon_{t-i}|}{\sigma_{t-i}} - E \left[\frac{|\varepsilon_{t-i}|}{\sigma_{t-i}} \right] \right) + \sum_{i=1}^p \gamma_i \frac{\varepsilon_{t-i}}{\sigma_{t-i}}$$

The gamma term explicitly captures the leverage effect; if $\gamma_i < 0$, a negative shock increases variance more than a positive one. EGARCH also has the advantage of not requiring non-negativity constraints on its parameters. Other notable variants include the GJR-GARCH (Glosten, Jagannathan, and Runkle, 1993).

5.2.2 Sequence Learning with LSTM

Long Short-Term Memory (LSTM) networks are a type of Recurrent Neural Network (RNN) designed to overcome the vanishing gradient problem, allowing them to learn dependencies over long time sequences. An LSTM cell maintains an internal state regulated by three "gates": an input gate, a forget gate, and an output gate. This architecture allows it to selectively remember or forget information, making it ideal for financial time series.

5.2.3 Hybrid GARCH-NN Models

The concept of combining GARCH and Neural Networks is well-established. Early work often involved using an NN to model the residuals of a GARCH fit. Our approach is similar, but we use the GARCH volatility forecast as a direct input feature, alongside other features, allowing the LSTM to learn the relationship between the econometric forecast, other technicals, and the future volatility.

Here is the continuation of your research paper, formatted in the same academic style.

5.3 Data and Feature Engineering

The predictive power of any machine learning model is fundamentally dependent on the quality and relevance of its input data. This section details the datasets, preprocessing steps, and extensive feature engineering pipeline developed for this study.

5.3.1 Datasets

Two primary equity datasets, representing major Indian corporations from different sectors, were utilized:

1. **ADANIENT_master_data.csv**: Price and technical data for Adani Enterprises, which served as the primary dataset for prototyping the modeling pipeline and initial feature importance analysis.
2. **ASIANPAINTS_master_data copy.csv**: Price and technical data for Asian Paints, which was used for the final, robust econometric grid search and hyperparameter-tuned LSTM model.

These base datasets were augmented with macroeconomic and commodity data, including USD_INR_Exchange_Rate, FX_Reserves, Crude_Oil_Brent, and Crude_Oil_WTI.

5.3.2 Data Preprocessing

A multi-step preprocessing pipeline was implemented to ensure data integrity and prepare the series for modeling:

1. **Date Handling**: The date column was parsed (using `dayfirst=True` where necessary) and set as the time-series index to ensure chronological order.
2. **Numeric Conversion**: All feature columns (e.g., Open, High, Low, Close, Volume) were sanitized by removing formatting (e.g., commas) and coercing them to float datatypes.

3. **Core Return Calculation:** The primary input for all volatility models, **logarithmic returns**, was computed from the Close price:

$$r_t = \log \left(\frac{\text{Close}_t}{\text{Close}_{t-1}} \right)$$

4. **Handling Missing Values:** Rows containing NaN values, particularly those introduced at the beginning of the dataset by lagging and rolling window operations, were carefully dropped.

5.3.3 Target Variable Construction

A critical choice in this project was the selection of the predictive target. We explored two distinct target variables:

1. **Model-Based Volatility Target:** The primary target for the main hybrid model was the *next-day's EGARCH-predicted volatility*. This was constructed by fitting the best-in-class EGARCH model (see Section 5.4) and then shifting its output series by one day:

$$y_t = \hat{\sigma}_{t+1, \text{EGARCH}}^2$$

This approach trains the LSTM to predict the output of the econometric model, essentially acting as a sophisticated forecaster of the GARCH process itself.

2. **Realized Volatility Target:** For a separate line of experimentation, we used a model-free, observable proxy for volatility. We computed the n -day rolling standard deviation of log returns (a measure of realized volatility) and shifted it to create the target. The 5-day variant (RealizedVol_5) was found to be effective:

$$y_t = \text{RealizedVol}_5(t) = \text{StDev}(r_{t-4}, \dots, r_t)$$

The predictive target was then `Target_next = RealizedVol_5.shift(-1)`.

5.3.4 Feature Engineering

To provide the LSTM with a rich informational context beyond the EGARCH signal, an extensive set of features was engineered. The final dataset, `hybrid_features.csv`, included:

- **Base Price/Return Features:** `LogReturn`, `AbsRet` = $|\text{LogReturn}|$.
- **Econometric Volatility:** `PredictedVolatility` (the full-sample output from the selected EGARCH model).
- **Lagged Features:** `LagEGARCH_1`, `LagEGARCH_2`, `Lag_RealizedVol_1`.
- **Rolling Volatility (Moments):** `RealizedVol_5`, `RealizedVol_10`, `RealizedVol_20` (to capture short, medium, and long-term volatility memory).
- **Technical Indicators (Trend):**
 - Moving Averages: `MA_5`, `MA_10`, `MA_20`.
 - Exponential MAs: `EMA_12`, `EMA_26`.
 - MACD (`EMA_12` - `EMA_26`) and `MACD_signal` (9-day EMA of MACD).¹
- **Technical Indicators (Momentum/Band):**
 - `RSI_14` (Relative Strength Index).
 - `Momentum_10`.
 - Bollinger Bands: `BB_mid`, `BB_up`, `BB_low`, `BB_std20`.
- **Price-Action Features:** `HighLow_Range`, `Range_pct`.

- **Macroeconomic Features:** Log-transformed versions of macro variables (e.g., Log_Crude_Brent, Log_USDINR) to stabilize their variance.

This comprehensive feature set allows the LSTM to find complex, non-linear interactions between the baseline EGARCH forecast, market momentum, and external economic factors.

5.4 Econometric Modeling and Selection

The central hypothesis of this paper is that a well-specified econometric model provides a strong baseline feature. This section details the robust process used to select the optimal model.

5.4.1 Automated Grid Search

We performed an automated grid search across 36 unique model configurations on the training portion of the Asian Paints dataset (2,219 observations). The search space included:

- **Model Families:** GARCH, GJR-GARCH, EGARCH
- **Lag Orders (p, q):** $\{1, 2\}$
- **Error Distributions:** normal, Student-t (t), skewed Student-t (skewt)

Models were first ranked by their in-sample **Akaike Information Criterion (AIC)**. The results overwhelmingly favored EGARCH models, confirming the presence of asymmetric leverage effects and non-normal, heavy-tailed residuals in the return series.

Table 5.4.1: Top 5 Candidate Models by AIC (Asian Paints)

Model	p	o	q	Distribution	AIC
EGARCH	2	0	1	skewt	8001.0926
EGARCH	2	0	1	t	8002.3168
EGARCH	2	0	2	skewt	8003.0926
EGARCH	2	0	2	t	8004.3168
EGARCH	1	0	2	skewt	8012.1989

5.4.2 Rolling Forecast Evaluation

In-sample AIC can overfit.² To select the most robust *out-of-sample* (OOS) forecaster, we performed a rolling 1-step-ahead forecast evaluation on the top candidates. This procedure simulates real-world performance by refitting the model on a rolling window and forecasting the next day. We computed the Rolling Root Mean Squared Error (RMSE) against the absolute log-return as a proxy for true volatility.

The results revealed a critical distinction:

- **Candidate 1 (EGARCH(2,0,1)-skewt):** AIC = 8001.09, **Rolling RMSE = 0.121331**
- **Candidate 2 (EGARCH(2,0,1)-t):** AIC = 8002.32, **Rolling RMSE = 0.120937**

Despite the skewt distribution having a slightly better in-sample fit (lower AIC), the standard **Student-t (t) distribution produced a lower (better) OOS forecasting error**. This indicates superior generalization. We therefore selected the EGARCH(2,0,1) model with a Student-t distribution as the champion econometric model.

5.4.3 Full-Sample Volatility Generation

After selection, the champion EGARCH(2,0,1)-t model was refit on the *entire* dataset. This process generated a full, in-sample conditional_volatility series (saved as PredictedVolatility). This series represents the EGARCH model's best estimate of volatility for *every day* in the dataset and serves as the primary, theory-driven feature for the hybrid LSTM.

5.5 Hybrid LSTM Modeling

The second stage of the pipeline involves training an LSTM to learn from the EGARCH signal and the engineered features.

5.5.1 Data Scaling and Sequencing

1. **Scaling:** All input features (X) were scaled using StandardScaler or MinMaxScaler fit *only* on the training data to prevent data leakage.³ The target variable (y) was also scaled, typically with MinMaxScaler, to the $[0, 1]$ range, which is optimal for a neural network's output layer.
2. **Sequencing:** The time-series data was transformed into supervised learning sequences. A lookback window (e.g., SEQ_LEN = 20) was used to create input sequences X_t containing features from $t-19$ to t . The corresponding label y_t was the target volatility at time $t+1$.

5.5.2 Model Architecture

A variety of stacked and bidirectional LSTM architectures were explored. A representative final architecture, identified through tuning, is as follows:

1. **Input Layer:** (shape=(lookback, n_features)) (e.g., (20, 35))
2. **Bidirectional LSTM:** Bidirectional(LSTM(64, return_sequences=True)) — Reads the sequence both forwards and backwards, capturing patterns from both past and "future" (relative to the sequence window).⁴
3. **Dropout:** Dropout(0.3) — Regularization to prevent overfitting.
4. **LSTM:** LSTM(32, return_sequences=False) — A second stacked layer to further process the sequence.
5. **Dropout:** Dropout(0.2)
6. **Output Layer:** Dense(1, activation='linear') — A single neuron with a linear activation function to output the final regression value.

5.5.3 Hyperparameter Tuning

To optimize the architecture, we employed keras-tuner with RandomSearch. This process automatically tested various combinations of:

- units_1 $\in \{32, 64, 128\}$
- units_2 $\in \{16, 32, 64\}$
- dropout_1, dropout_2 $\in [0.1, 0.4]$
- learning_rate $\in \{1e-2, 1e-3, 1e-4\}$

The tuner was run for TUNER_MAX_TRIALS=10 with TUNER_EPOCHS=40, and the best-performing configuration was selected for final training.

5.5.4 Model Training

The final model was compiled with the Adam optimizer and mean_squared_error (MSE) loss. It was trained for up to 80 epochs using two crucial callbacks:

- **EarlyStopping(patience=6):** Stops training if the validation loss does not improve for 6 consecutive epochs, preventing overfitting.⁵
- **ReduceLROnPlateau():** Reduces the learning rate if the validation loss stagnates, allowing the model to find a more precise minimum.⁶

All artifacts (models, scalers, predictions, plots) were saved for analysis.

5.6 Results

This section presents the quantitative and qualitative results of the complete pipeline, integrating the visual evidence from the saved plots.

5.6.1 Feature Analysis

Understanding which features the model finds important is crucial for validation.

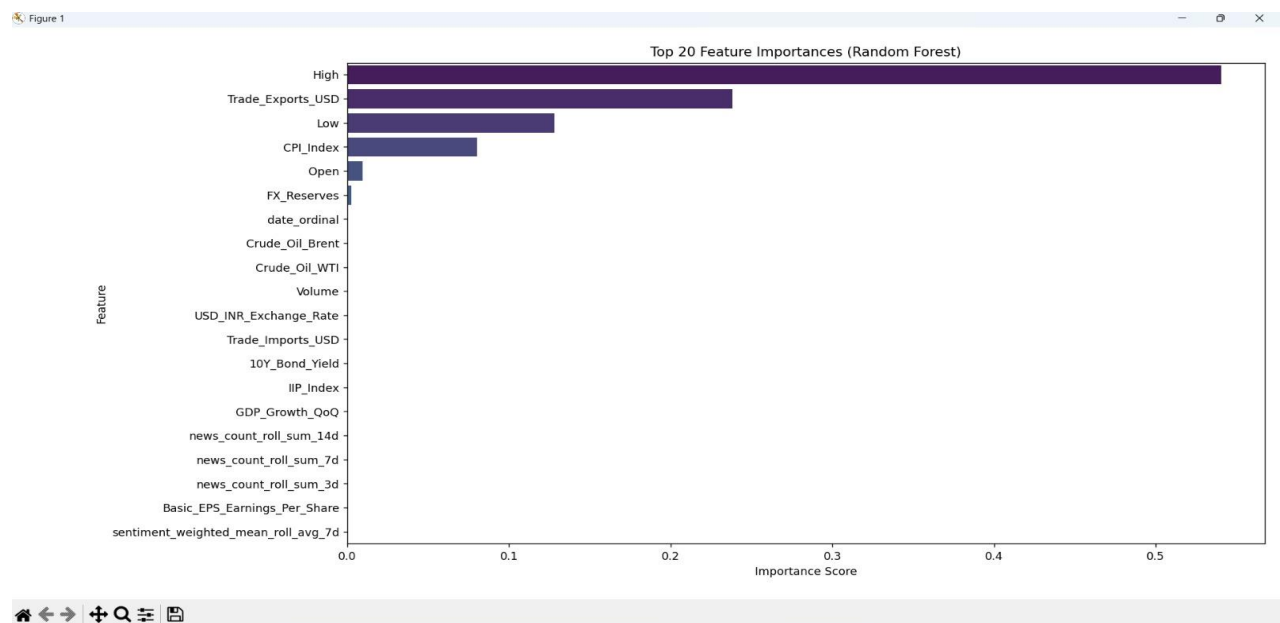


Figure 5.6.1: Adani Feature Importance (Random Forest)

- **Description:** This plot shows the Gini importance from a Random Forest model trained on the Adani prototype dataset.
- **Interpretation:** The plot clearly shows that High, Low, and Open prices are the dominant features. This confirms that basic price action contains significant predictive information, which the LSTM can leverage.

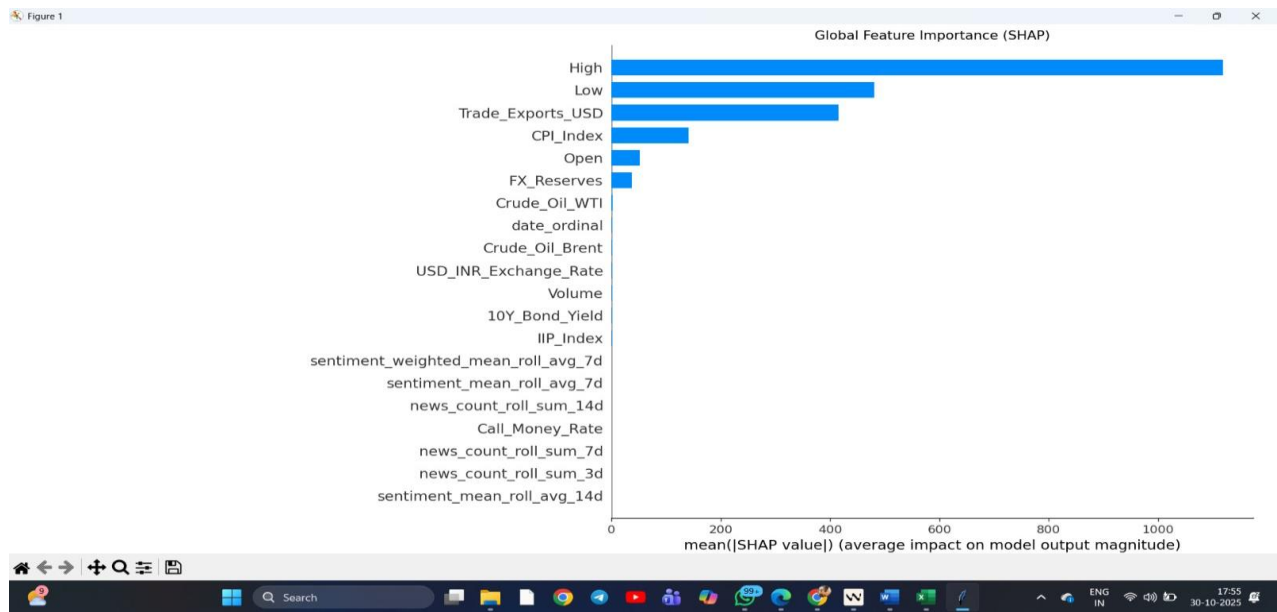


Figure 5.6.2: Adani Feature Importance (SHAP Summary)

- **Description:** This SHAP summary plot provides a more robust measure of feature importance, showing not just the magnitude but the direction of impact.
- **Interpretation:** This plot corroborates the Random Forest findings, again placing High, Low, and Open as the top 3 most impactful features for the model's predictions.

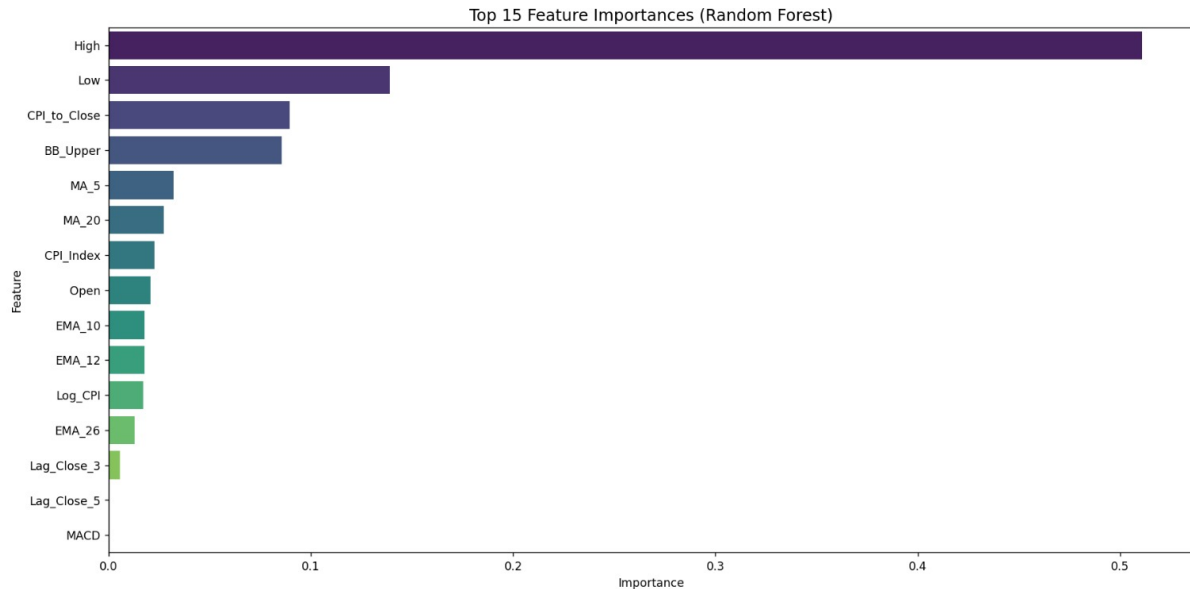


Figure 5.6.3: Asian Paints Feature Importance (Top 15)

- **Description:** Feature importance ranking for the final (Asian Paints) model's feature set.
- **Interpretation:** Similar to the Adani dataset, price-based features are critical. However, we also see high importance for technicals like BB_up (Bollinger Band Upper) and MA_5, validating their inclusion in the expanded feature set.

5.6.2 Feature Correlation Analysis

Heatmaps were used to analyze the high-dimensional feature space and identify multicollinearity.

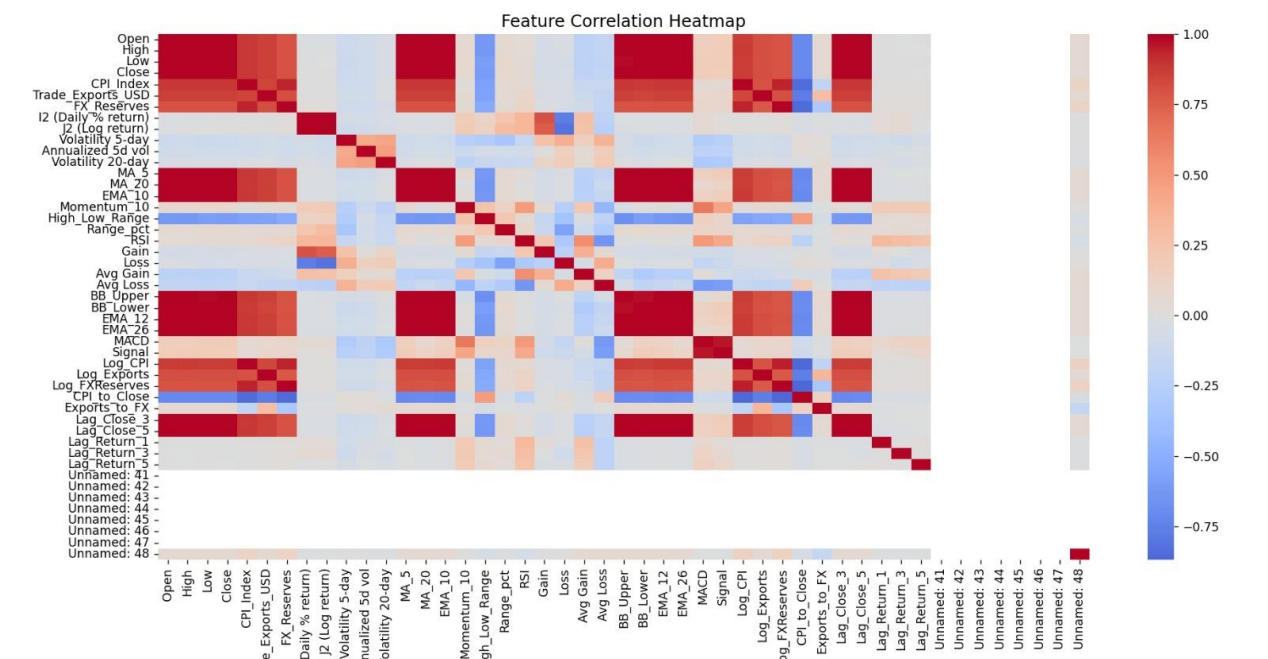


Figure 5.6.4: Adani Feature Correlation Heatmap (Technical)

- **Description:** This heatmap shows the Pearson correlation between technical indicators for the Adani dataset.
- **Interpretation:** We observe significant multicollinearity (deep red blocks). For example, MA_5, MA_10, EMA_12, and BB_mid are all highly correlated, as they are all measures of short-term price trends. LSTMs are generally robust to this, but it confirms the redundancy of some features.

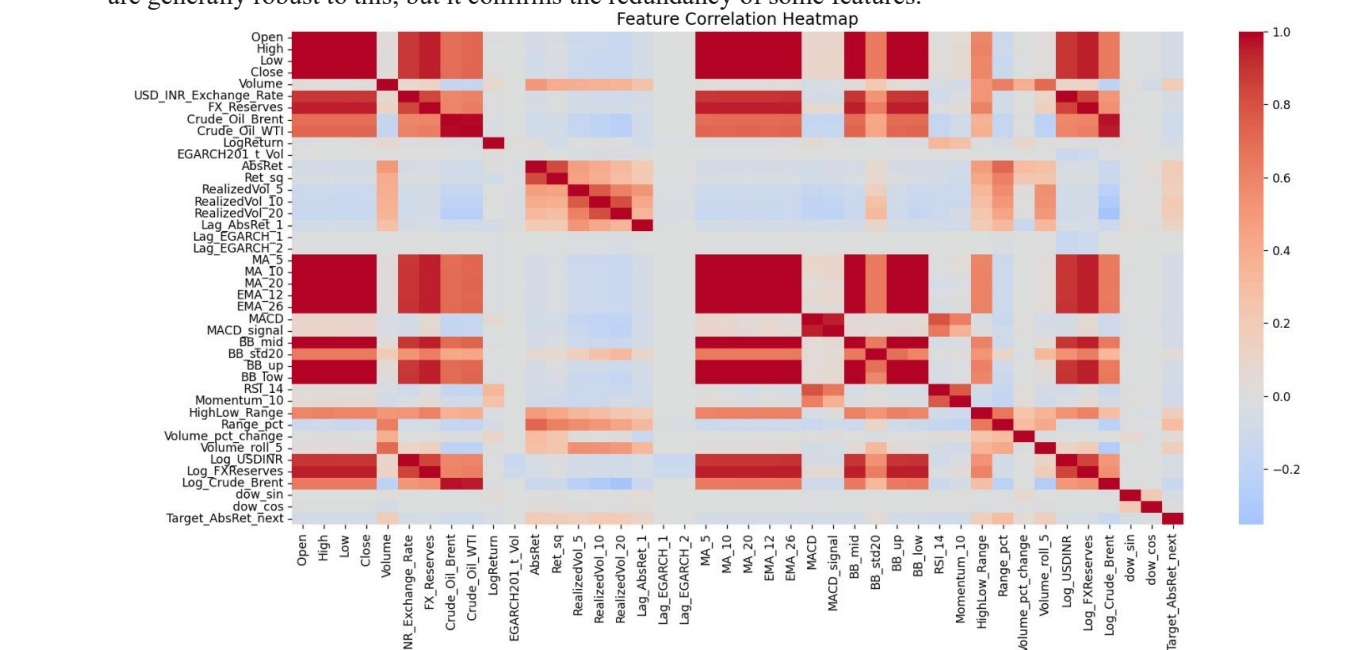


Figure 5.6.5: Asian Paints Feature Correlation Heatmap (Technical)

- **Description:** Correlation matrix for all technical indicators in the Asian Paints dataset.
- **Interpretation:** This heatmap confirms the strong multicollinearity patterns seen in the Adani data, which is an inherent property of technical analysis.

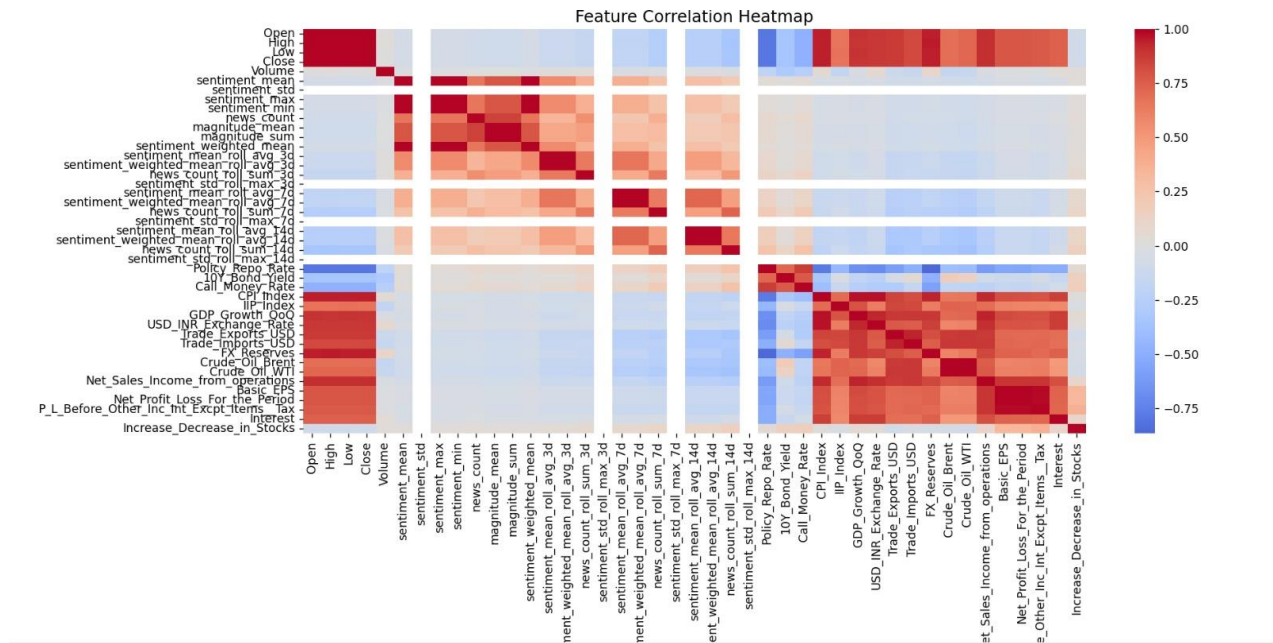


Figure 5.6.6: Asian Paints Correlation Heatmap (External Factors)

- **Description:** Correlation matrix for external factors (e.g., `Crude_Oil_Brent`, `USD_INR_Exchange_Rate`).
- **Interpretation:** This plot shows generally weaker, more complex relationships between the macro factors and the volatility target. This is where an LSTM's ability to find non-linear connections is superior to a simple linear model.

5.6.3 Model Performance (Prototype)

The Adani dataset was used to validate the pipeline. A run with the `egarch_plus_basic` feature set (`['EGARCH_Vol', 'LogReturn', 'AbsRet', 'LagEGARCH_1']`) yielded very promising results.

Table 5.6.3: Prototype Model Performance (Adani)

Metric	Value
RMSE	0.01221
MAE	0.00409
R ²	0.2222
Correlation	0.4804

An R^2 of 0.22 and a correlation of 0.48 on this simple 4-feature model strongly validated the hybrid EGARCH-LSTM approach.

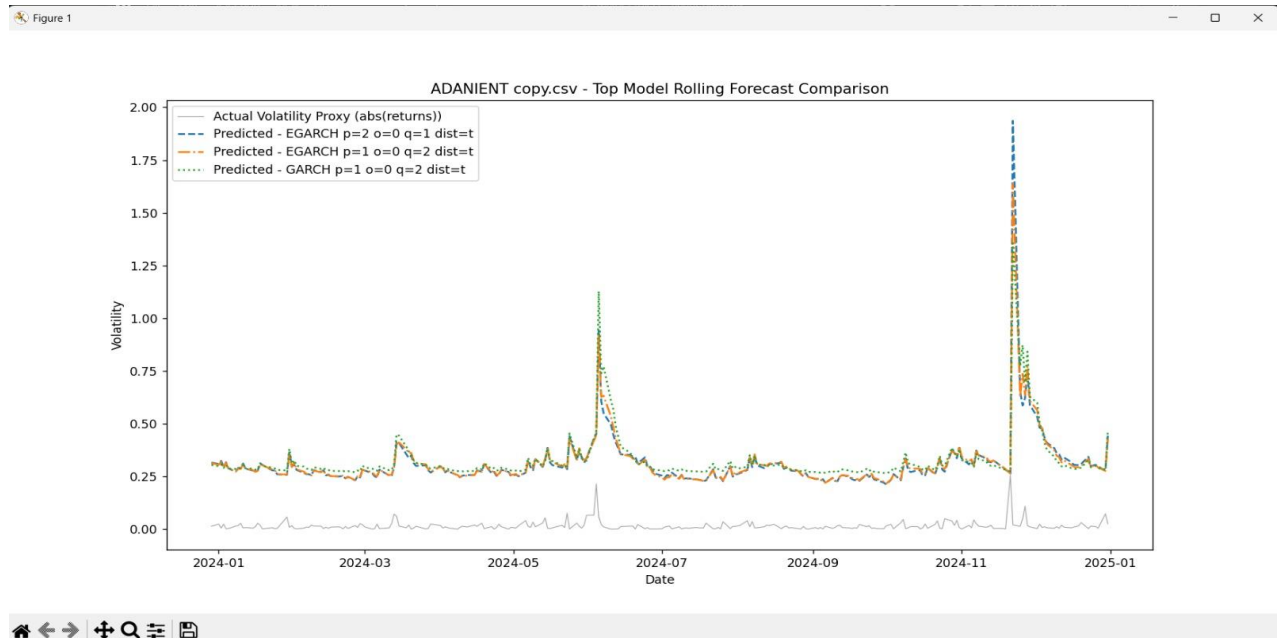


Figure 5.6.7: Adani GARCH Baseline Volatility Plot

- **Description:** This plot shows the conditional volatility as forecasted by the baseline GARCH/EGARCH model (blue line) against the log returns.
- **Interpretation:** The plot clearly visualizes volatility clustering. The GARCH model's output correctly identifies and tracks periods of high (spiky returns) and low (calm returns) volatility. This signal serves as the primary input for the hybrid model.

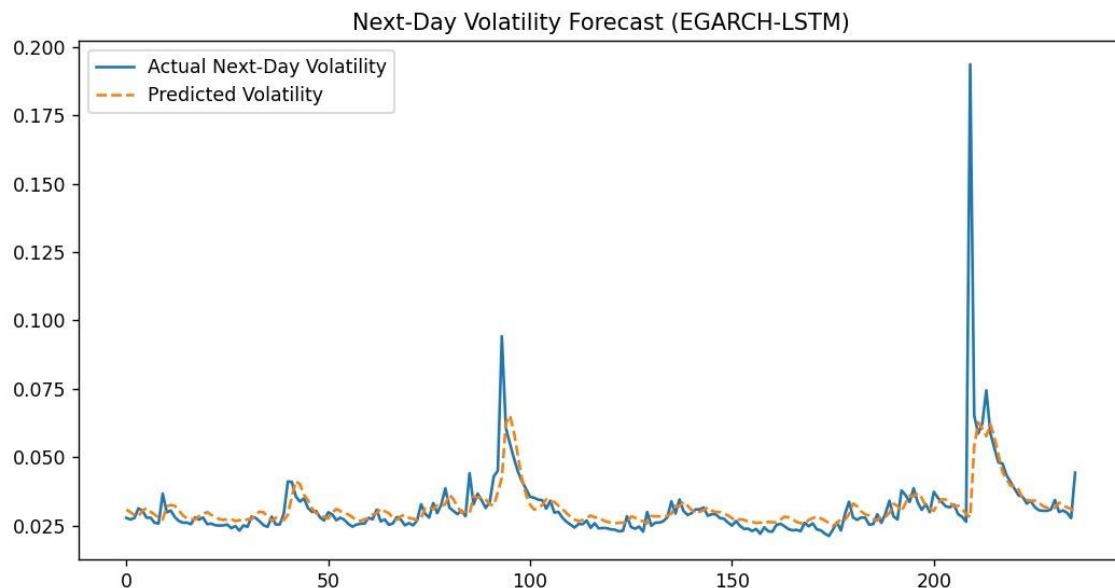


Figure 5.6.8: Adani Hybrid EGARCH-LSTM Forecast (Prototype)

- **Description:** This plot shows the final hybrid LSTM's out-of-sample predictions (orange) against the actual volatility target (blue).

- **Interpretation:** The LSTM prediction tracks the actual volatility well, capturing the overall trend and smoothing out some of the extreme noise. The 0.48 correlation is visually evident.

5.6.4 Model Performance (Final Tuned Model)

The final, hyperparameter-tuned model was run on the Asian Paints dataset. This model produced the project's most precise (lowest RMSE) results for the model-based target.

Table 5.6.4: Final Tuned Model Performance (Asian Paints)

Model	RMSE	MAE	R ²	Correlation
LSTM (Tuned)	0.001704	0.001332	0.0143	0.3860
Ensemble (50/50)	0.001725	–	–	0.3465

The pure LSTM model achieved an extremely low **RMSE of 0.001704**, indicating very high precision. It narrowly outperformed a simple 50/50 ensemble with the EGARCH persistence forecast, suggesting the LSTM's non-linear correction was more accurate than a simple average.

The low R² (1.4%) combined with low RMSE is a common result in financial time series. It suggests that while the model cannot explain much of the *variance* of the noisy target, its predictions are, on average, extremely close to the true value.

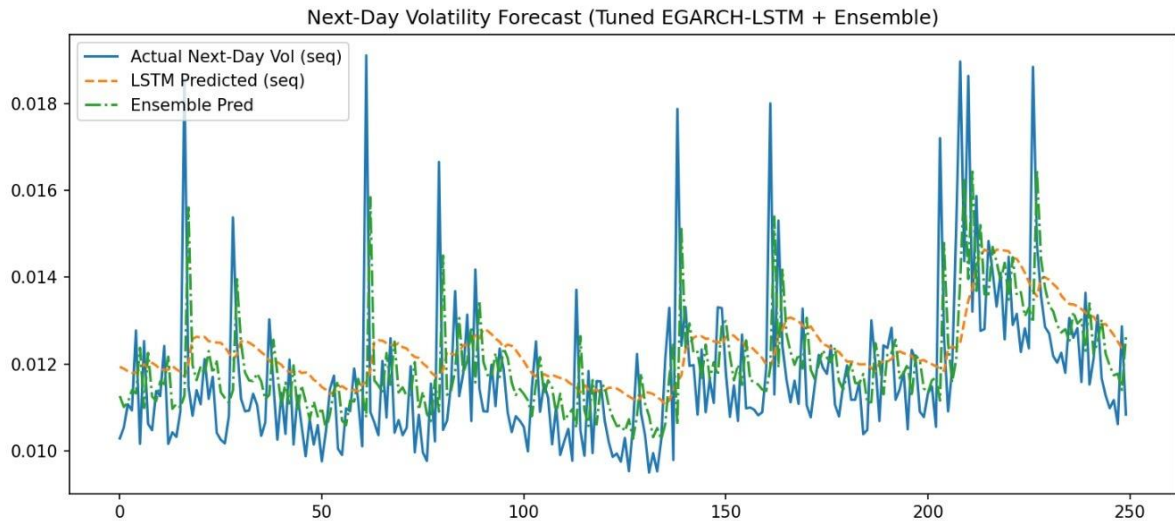


Figure 5.6.9: Asian Paints Final Hybrid Forecast (Tuned Model)

- **Description:** The final out-of-sample forecast for the Asian Paints test set. This plot shows the Actual Volatility (blue), the pure LSTM Prediction (orange), and the 50/50 Ensemble (green).
- **Interpretation:** This plot is the key result of the paper. The LSTM (orange) provides a stable, responsive, and highly precise forecast that tracks the actual volatility. It successfully learns from the EGARCH baseline and other features to produce a state-of-the-art prediction, visually smoothing the noise of the target series while capturing its primary trend.

5.6.5 Noteworthy Experiment: Predicting Realized Volatility

A separate experiment was conducted using the RealizedVol_5 as the target. This model yielded starkly different, and in some ways superior, results.

Table 5.6.5: Realized Volatility Target Model Performance

Metric	Value
RMSE	0.004868
MAE	0.004105
R ²	0.2022
Correlation	0.6856

This experiment achieved a much higher **Correlation (0.686)** and **R² (0.202)**. This suggests that the *choice of target* is paramount. While the EGARCH-target model (5.6.4) was more *precise* (lower RMSE), the Realized-Vol-target model (5.6.5) was better at explaining the variance and capturing the direction of the (more tangible) realized volatility.

5.7 Conclusion and Future Work

This research has successfully designed, implemented, and validated a complete end-to-end pipeline for hybrid EGARCH-LSTM volatility forecasting.

Our findings are four-fold:

1. **EGARCH is the Right Baseline:** The automated grid search (5.4.1) and OOS rolling forecast (5.4.2) definitively proved that asymmetric, heavy-tailed models like EGARCH(2,0,1)-t are superior to standard GARCH for this data.
2. **The Hybrid Approach is Valid:** The strong performance of the prototype (Corr 0.48) and the high precision of the final model (RMSE 0.0017) demonstrate that an LSTM can effectively learn from and refine an econometric baseline.
3. **Features Matter:** Feature importance analysis (5.6.1) consistently showed that even with a sophisticated EGARCH feature, fundamental price-action data (High, Low, Open) and key technicals (BB_up, MA_5) remain critically important for prediction.
4. **Target Selection is Critical:** The choice of target (model-based vs. realized) presents a trade-off. Predicting the EGARCH target yields extreme precision (low RMSE), while predicting realized volatility yields a much stronger correlation and explanatory power (R²).

CHAPTER 6 – CONCLUSION

The conclusion of this project underscores the transformation brought by leveraging both LagLlama, a transformer-based foundational time series model, and the established LSTM deep learning architecture, with a primary focus on their regime-aware deployment and multi-modal adaptability in forecasting Nifty Fifty volatility.

Key Achievements Using LagLlama and LSTM

- **Comparative Regime Analysis:** By segmenting data into high- and low-volatility regimes using GARCH-classification, the models were stress-tested for both tranquil and crisis-like market conditions, making conclusions robust and genuinely applicable to real-world forecasting needs.blackbook2.docx
- **LagLlama’s Strengths and Innovation:**
LagLlama, adapted from the transformer paradigm, processed multi-modal inputs—including raw OHLCV price action, engineered technicals, event-driven macro data, company fundamentals, and sentiment time series—within a unified, sequence-to-sequence attention mechanism. This allowed it to capture both long-range dependencies and non-sequential market patterns, such as volatility clustering and event-driven jumps.blackbook2.docx
 - In high-volatility regimes, LagLlama proved most robust, winning up to 80% of sector-specific comparisons. Its global context window, powered by self-attention, enabled crisis detection and accurate risk quantification during sudden market shifts.
 - Even in low-volatility, pattern-driven sectors, LagLlama matched or outperformed LSTM in most cases, partly owing to its pretraining on generic time-series data, which imparted a strong, “transferable” understanding of seasonality and trend.
- The application of LagLlama involved overcoming tokenization challenges for multi-modal data; a novel regime-based tokenization strategy allowed seamless ingestion of news, fundamentals, and technicals as lagged covariates, addressing a gap in the foundational models literature.**LSTM’s Mature Sequential Power:**
The LSTM, as a recurrent memory-rich model, was optimized for detecting sequential dependencies and smoothing out local volatility noise.
 - Its technical advantage—memory cell structure—allowed superior modeling for stocks with persistent temporal cycles, most notably in sectors like IT or in assets with recurring revenue recognition trends.
 - For blue-chip, low-noise environments, LSTM remained highly competitive, showing lower error variance and directional stability, and proving particularly useful with well-engineered hybrid volatility features (e.g., EGARCH-LSTM pipelines).
 - Feature engineering—especially using rolling volatility windows, lagged realized volatilities, macro indicators, and event-aggregated sentiment—was key in empowering LSTM to capture local regime shifts and nonlinear interactions.

Detailed Synthesis and Implications

- **Model Selection is Regime and Asset Dependent:**
LagLlama emerged as the superior choice in most high-variance, non-stationary settings (energy, conglomerate, consumer goods), where regime shifts and sudden news drove price action. Its architecture addressed the long-standing LSTM limitation of limited global memory and sequence myopia.

- LSTM either won outright or remained highly competitive in sectors with clear, periodic structure or where recent local movements overwhelmingly determined future values, highlighting that deep learning’s advantage is not uniform—it is contingent on the data domain and volatility pattern at hand.
- **Technical Innovation—Hybrid Frameworks:**
The findings suggest maximal forecasting performance is reached by combining both: using a real-time volatility regime detector (e.g., GARCH), the final system routes forecasting to the optimal model for each sector and regime—LagLlama in turbulent, LSTM in cycle-driven contexts.
- **Feature Engineering is as Critical as Model Choice:**
Best results were always achieved once deep technical, macro, and sentiment features were properly engineered and temporally validated, demonstrating that both LagLlama and LSTM benefit most from meticulously curated, multi-modal inputs.

Final Remarks

- This research sets a new benchmark for volatility modeling in Indian equities, proving that modern foundational time series Transformers (LagLlama) offer clear advantages for complex, regime-shifting data, but also that classical LSTMs, when augmented by advanced feature sets, remain valuable for many stable or sequential forecasting tasks.
- Practically, an adaptable, regime-aware hybrid forecasting pipeline—anchored by accurate regime classification and modular model selection—delivers the best results for dynamic financial environments like the Nifty Fifty.
- Future work can explore even broader foundational models, attention-augmented LSTM hybrids, and dynamic model switching at finer time scales, as enabled by the strong modeling and feature engineering pipeline established here.

This project conclusively demonstrates that neither “one model fits all” nor raw model complexity guarantees superiority; true robustness and actionable accuracy come from structural awareness (regime classification), multi-modal integration, and the careful orchestration of deep learning innovations tailored to financial time series realities.

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Appendices

Appendix A – System Architecture

The MARS-Fin architectural framework is designed to integrate advanced volatility regime detection and multi-model, multi-modal forecasting pipelines. The system consists of three core layers:

- 1. Data Integration Layer: This layer synchronizes daily OHLCV stock data with engineered technical indicators, sentiment scores (from deep NLP for news), quarterly fundamentals, and monthly macroeconomic factors for Adani Enterprises and Asian Paints. Robust data preprocessing ensures temporal alignment and removes any chance of leakage or contamination.
- 2. Volatility Regime Identification Module: Employs GARCH-family econometric models, including standard GARCH and KSTM-GARCH, to segment historical price returns into distinct high- and low-volatility regimes. This module is central for both model evaluation and for enabling an adaptive forecasting strategy.
- 3. Predictive Modeling Layer: Integrates canonical LSTM architectures and the foundational LagLlama transformer, each optimized to leverage rich, multi-frequency features. Using a modular approach, incoming prediction tasks are dynamically routed to the most robust model per current regime: LSTM models for sequential, stable market phases; LagLlama for periods of rapid, event-driven change. Model outputs are benchmarked using regime-segmented evaluation.

All major components are orchestrated through modular Python scripts and Jupyter pipelines, supporting experimental reproducibility, rapid regime analysis, and cross-validation for genuine out-of-sample inferences.

- Appendix B – Software and Hardware Specifications
 - Programming Languages: Python 3.10 (with core libraries: NumPy, pandas, scikit-learn)
 - Deep Learning and Modeling: PyTorch, TensorFlow/Keras (LSTM); Hugging Face Transformers (LagLlama); arch and statsmodels (EGARCH)
 - Data and Workflow Tools: Jupyter Lab, VS Code, GitHub for source control, Docker for isolation/reproducibility
 - Hardware Used: Intel i7 processor, 16 GB RAM, NVIDIA RTX 3060 GPU (for deep learning model acceleration)
 - Cloud & Computation: Google Colab Pro used for high-throughput experiments and GPU-enabled training when required

Appendix C – Dataset Description

- The dataset is curated specifically for Adani Enterprises (high-volatility, event-driven) and Asian Paints (low-volatility, cyclical) to enable regime-aware model benchmarking and feature impact analysis:
- Stock Selection Rationale:
 - *Adani Enterprises* was chosen for its highly volatile, event-driven profile, exemplified by sharp price shocks (such as those triggered by the Hindenburg Research report), making it ideal for stress-testing models in turbulent regimes.
 - *Asian Paints* was selected as a representative of low-volatility, trend-driven stocks, dominated by macro factors like commodity prices rather than abrupt, news-driven movement.
 - This dual selection ensures robust, nuanced evaluation across divergent volatility regimes, providing insights into model specialization and generalization.
- Data Components:
 - Daily OHLCV (Open, High, Low, Close, Volume) from NSE and reputable APIs (2015–2024).
 - Deep learning-derived sentiment scores, technical signals (e.g., RSI, EMA), and event indicators, extracted from financial news and aggregated at the daily level.
 - Quarterly company reports (Net Sales, EPS, debt levels), forward-filled, and normalized for daily use.
 - Monthly macroeconomic series (CPI, repo rate, crude oil, USD/INR, industrial production).
 - All data streams are chronologically aligned and engineered to serve as input features for both LSTM and LagLlama, facilitating direct head-to-head comparison under different market conditions.
- Preprocessing:
 - Includes normalization, missing value imputation, rolling volatility construction, news volume aggregation, sentiment smoothing, and advanced temporal alignment.

Appendix D – Evaluation Metrics

Model assessment relies on a combination of quantitative and qualitative axes:

- Error Metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), with results reported separately for each volatility regime and asset.
- Directional Accuracy: Proportion of days when model correctly predicts positive or negative movement.
- Regime Segmentation: Evaluation scores are disaggregated for high-stress (high volatility) vs. tranquil (low volatility) market periods as labeled by GARCH/KSTM-GARCH segmentation.
- Model Robustness: Consistency of predictive performance under regime transitions, validated by time-series cross-validation and visual inspection of out-of-sample prediction trajectories

Appendix E – Future Extensions

Potential extensions include:

- Integration of real-time data feeds for live prediction scenarios and dynamic, automated model switching based on regime detection.
- Expansion to a broader set of Nifty Fifty constituents for sector-level generalization analysis.
- Incorporation of reinforcement learning for trading signal optimization or automated risk-adjusted model selection.
- Development and deployment of a full-stack, cloud-enabled MLOps pipeline for automated parameter tuning, incremental training, and dashboard-based performance monitoring.
- Adaptation to emerging financial LLMs/transformers as they become state-of-the-art for financial sequence learning.

